

Penguins – Análisis Completo

Equipo 7

Integrantes:

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```
## [1] 344 8

## Rows: 344
## Columns: 8
## $ species      <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
## $ island       <chr> "Torgersen", "Torgersen", "Torgersen", "Torgersen", ~
## $ bill_length_mm <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
## $ bill_depth_mm <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
## $ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
## $ body_mass_g   <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
## $ sex           <chr> "male", "female", "female", NA, "female", "male", "f~
## $ year          <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~

##   species   island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
## 1 Adelie Torgersen      39.1          18.7           181          3750
## 2 Adelie Torgersen      39.5          17.4           186          3800
## 3 Adelie Torgersen      40.3          18.0           195          3250
## 4 Adelie Torgersen      NA            NA            NA            NA
## 5 Adelie Torgersen      36.7          19.3           193          3450
## 6 Adelie Torgersen      39.3          20.6           190          3650
##   sex year
## 1 male 2007
## 2 female 2007
## 3 female 2007
## 4 <NA> 2007
## 5 female 2007
## 6 male 2007

##   species  n
## 1 Adelie 152
## 2 Chinstrap 68
## 3 Gentoo 124

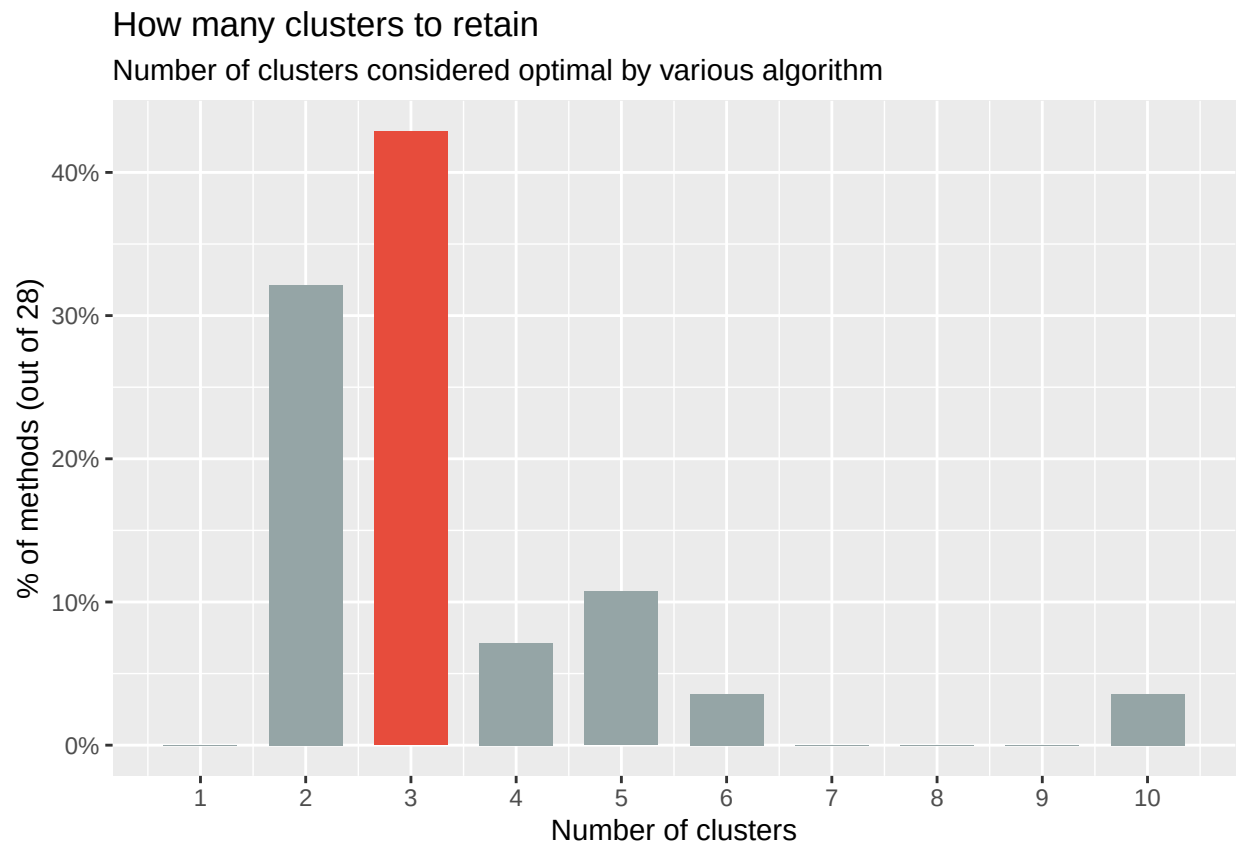
##
##   Adelie Chinstrap Gentoo
##   152          68      124
```

```
#explorando num clusters nc
```

```
## # Method Agreement Procedure:
```

```
##
```

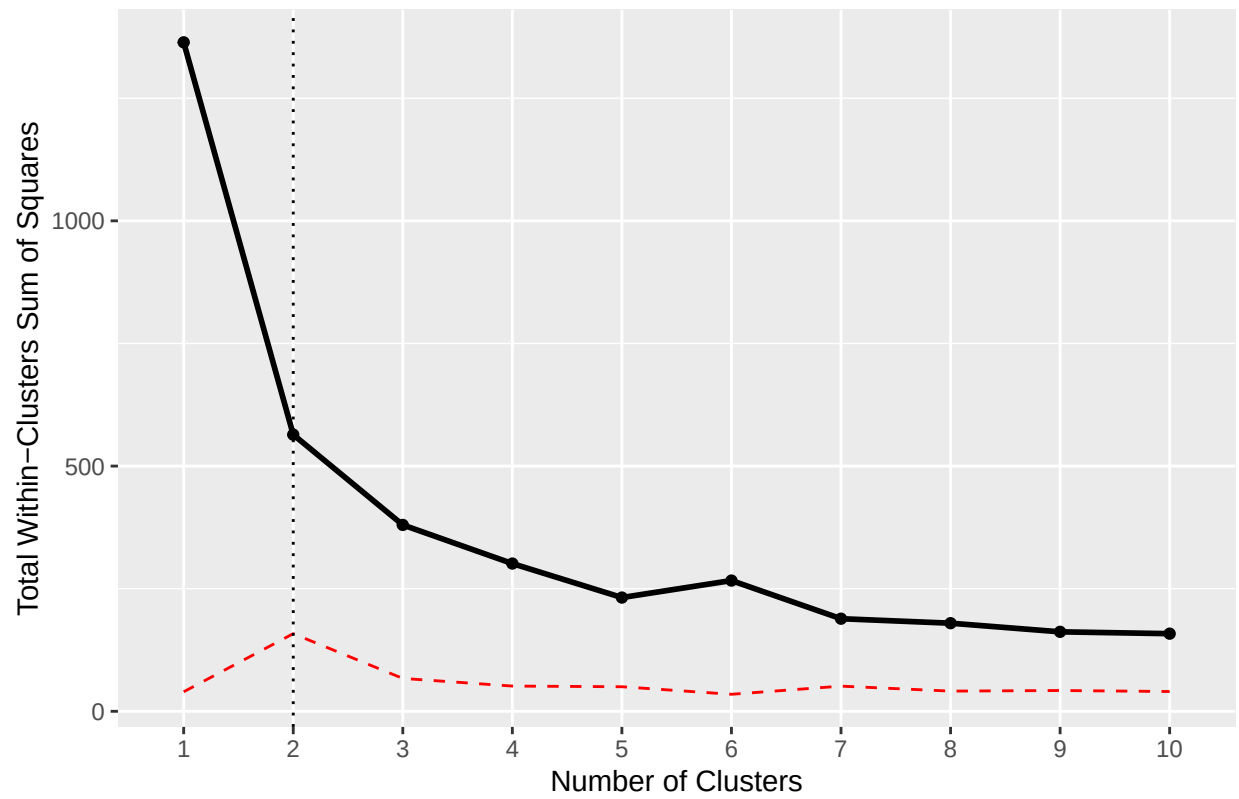
```
## The choice of 3 clusters is supported by 12 (42.86%) methods out of 28 (Hartigan, Scott, Marriot, tr
```



elbow

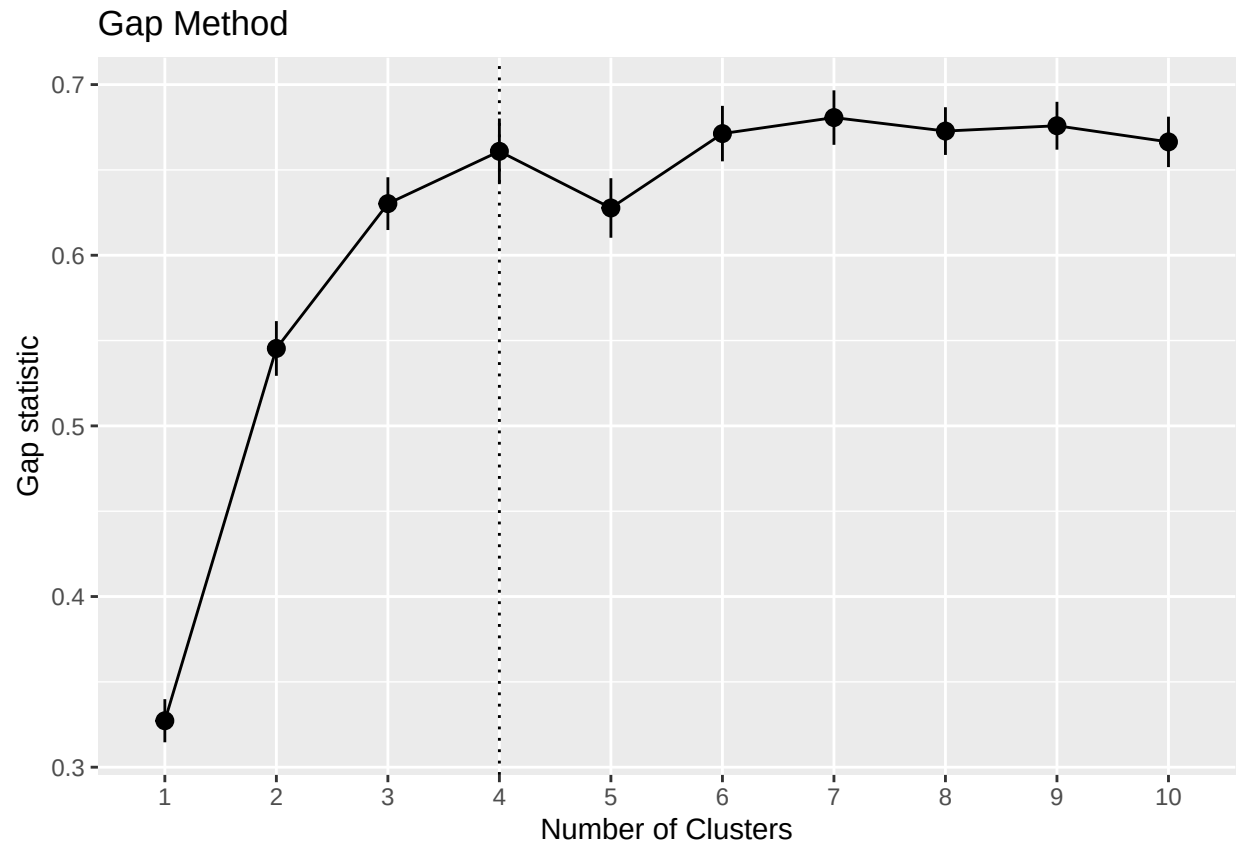
```
## The Elbow method, that aims at minimizing the total intra-cluster variation (i.e., the total within-
```

Elbow Method



GAP

The Gap method, that compares the total intracluster variation of k clusters with their expected value



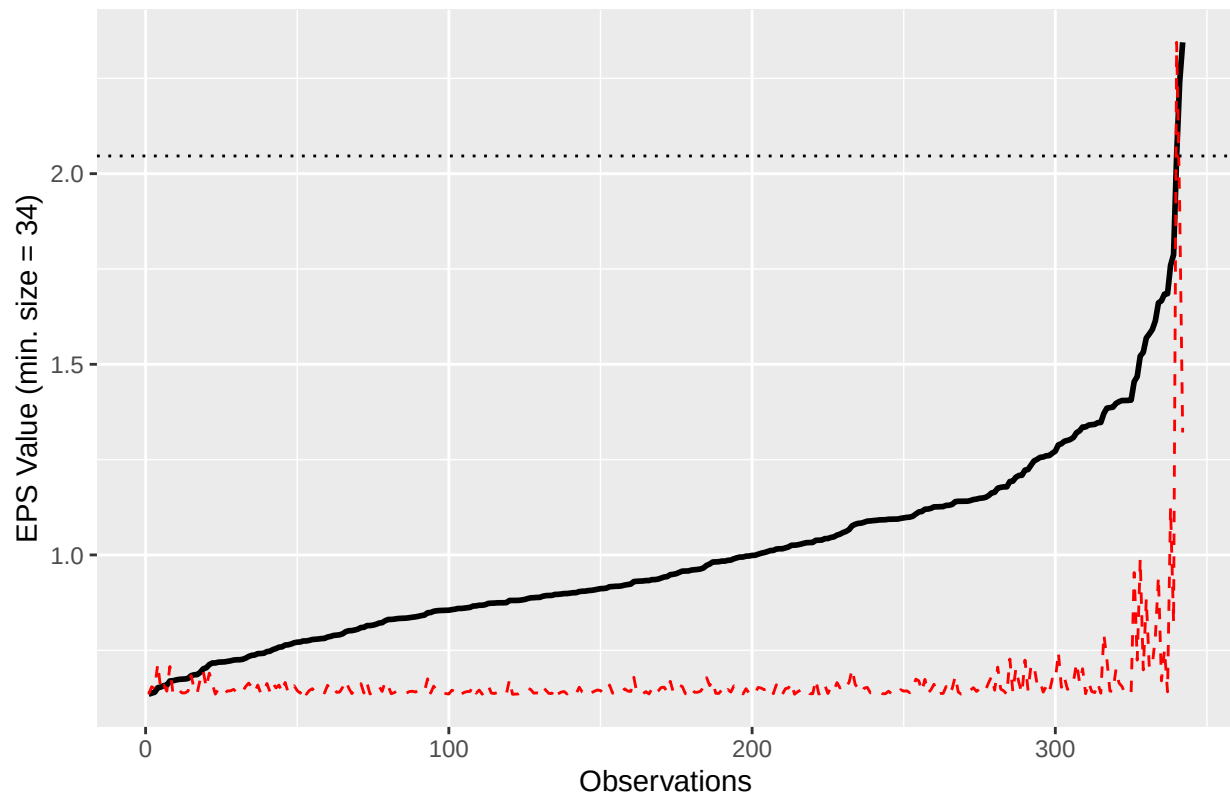
silhouette

The Silhouette method, based on the average quality of clustering, suggests that the optimal number of clusters is 4.

dbscan

The DBSCAN method, based on the total clusters sum of squares, suggests that the optimal eps = 2.046.

DBSCAN Method



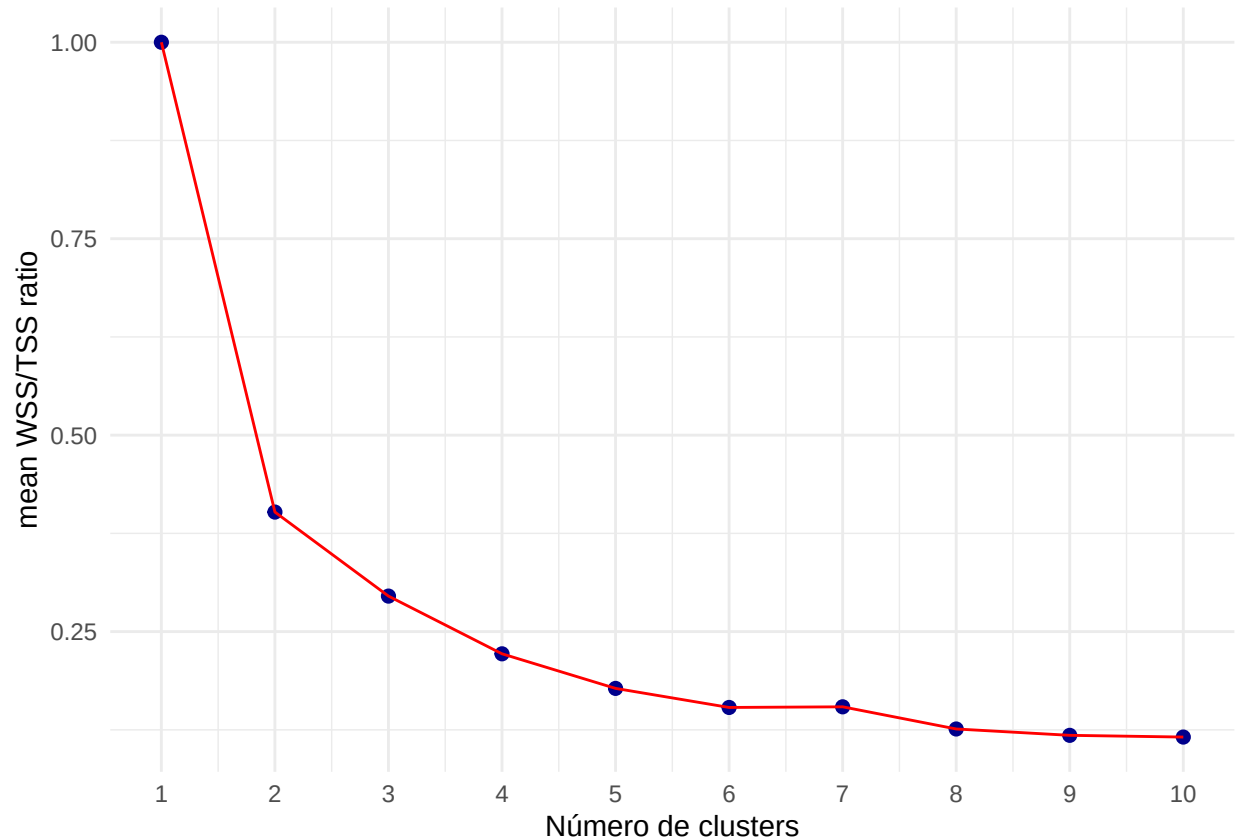
#K-Means

exploramos (kMeans)

```
## # A tibble: 30 x 7
##   num_clusters .metric      .estimator    mean    n std_err .config
##       <int> <chr>          <chr>      <dbl> <int>   <dbl> <chr>
## 1           1 sse_ratio      standard      1         10 0      Preprocesso~
## 2           1 sse_total      standard 1364         10 0      Preprocesso~
## 3           1 sse_within_total standard 1364         10 0      Preprocesso~
## 4           2 sse_ratio      standard  0.402         10 4.05e- 3 Preprocesso~
## 5           2 sse_total      standard 1364         10 0      Preprocesso~
## 6           2 sse_within_total standard  549.         10 5.52e+ 0 Preprocesso~
## 7           3 sse_ratio      standard  0.295         10 1.15e- 2 Preprocesso~
## 8           3 sse_total      standard 1364         10 0      Preprocesso~
## 9           3 sse_within_total standard  403.         10 1.56e+ 1 Preprocesso~
## 10          4 sse_ratio      standard  0.222         10 2.40e- 3 Preprocesso~
## 11          4 sse_total      standard 1364         10 0      Preprocesso~
## 12          4 sse_within_total standard  302.         10 3.28e+ 0 Preprocesso~
## 13          5 sse_ratio      standard  0.178         10 5.36e- 3 Preprocesso~
## 14          5 sse_total      standard 1364         10 0      Preprocesso~
## 15          5 sse_within_total standard  242.         10 7.32e+ 0 Preprocesso~
## 16          6 sse_ratio      standard  0.153         10 3.95e- 3 Preprocesso~
## 17          6 sse_total      standard 1364         10 0      Preprocesso~
## 18          6 sse_within_total standard  209.         10 5.38e+ 0 Preprocesso~
```

```
## 19      7 sse_ratio      standard      0.154      10 7.98e- 3 Preprocesso~
## 20      7 sse_total      standard    1364      10 0      Preprocesso~
## 21      7 sse_within_total standard    210.      10 1.09e+ 1 Preprocesso~
## 22      8 sse_ratio      standard      0.126      10 2.69e- 3 Preprocesso~
## 23      8 sse_total      standard    1364      10 0      Preprocesso~
## 24      8 sse_within_total standard    172.      10 3.67e+ 0 Preprocesso~
## 25      9 sse_ratio      standard      0.118      10 1.05e- 3 Preprocesso~
## 26      9 sse_total      standard    1364      10 0      Preprocesso~
## 27      9 sse_within_total standard    161.      10 1.43e+ 0 Preprocesso~
## 28     10 sse_ratio      standard      0.116      10 2.07e- 3 Preprocesso~
## 29     10 sse_total      standard    1364      10 2.40e-14 Preprocesso~
## 30     10 sse_within_total standard    158.      10 2.82e+ 0 Preprocesso~
```

```
## # A tibble: 1 x 2
##   num_clusters .config
##         <int> <chr>
## 1          10 Preprocessor1_Model10
```



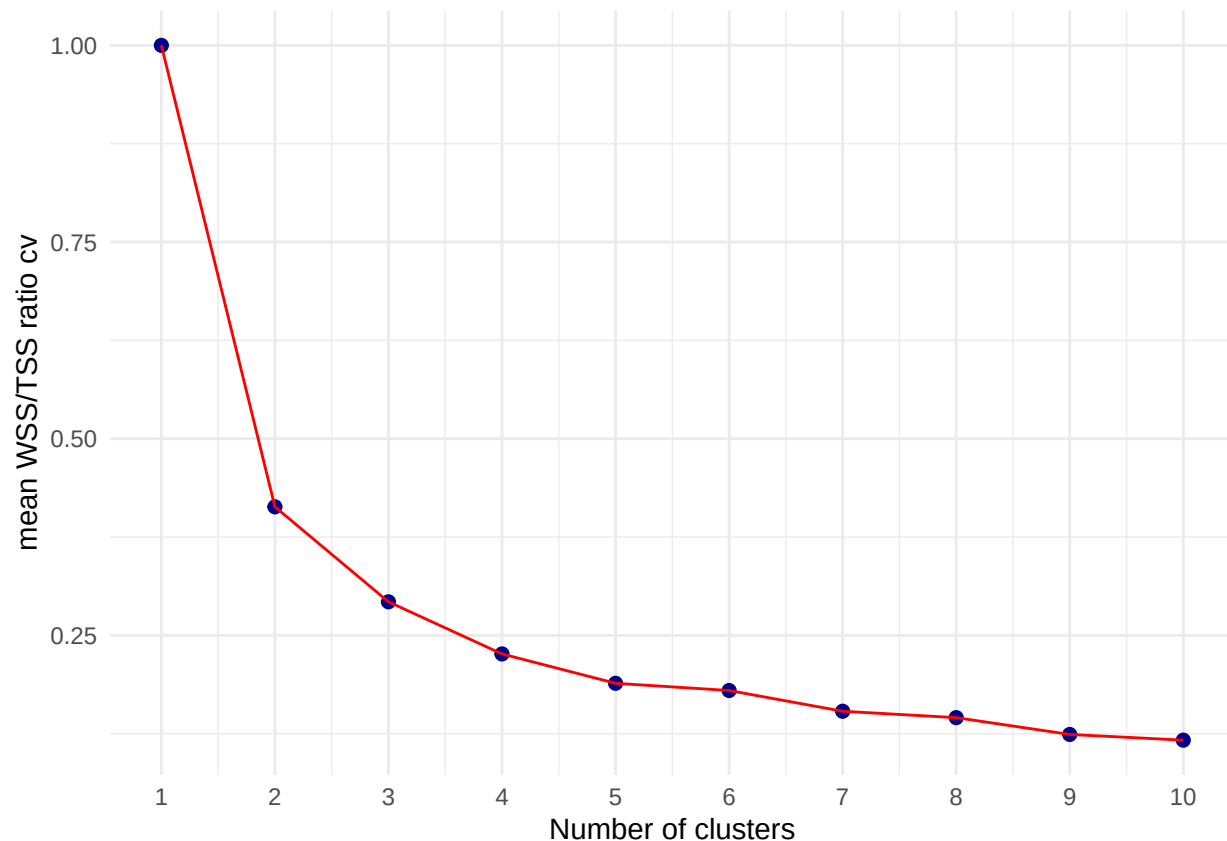
Validacion cruzada

```
## # A tibble: 10 x 1
##   num_clusters
##         <int>
## 1           1
```

```
## 2      2
## 3      3
## 4      4
## 5      5
## 6      6
## 7      7
## 8      8
## 9      9
## 10     10
```

```
## # A tibble: 30 x 7
##   num_clusters .metric      .estimator    mean     n std_err .config
##   <int> <chr>          <chr>      <dbl> <int>  <dbl> <chr>
## 1         1 sse_ratio      standard      1         10  0      Preprocesso~
## 2         1 sse_total      standard 1227.         10 0.533  Preprocesso~
## 3         1 sse_within_total standard 1227.         10 0.533  Preprocesso~
## 4         2 sse_ratio      standard   0.413        10 0.00190 Preprocesso~
## 5         2 sse_total      standard 1227.         10 0.533  Preprocesso~
## 6         2 sse_within_total standard   507.         10 2.42   Preprocesso~
## 7         3 sse_ratio      standard   0.293        10 0.0104  Preprocesso~
## 8         3 sse_total      standard 1227.         10 0.533  Preprocesso~
## 9         3 sse_within_total standard   359.         10 12.8    Preprocesso~
## 10        4 sse_ratio      standard   0.226        10 0.00411 Preprocesso~
## 11        4 sse_total      standard 1227.         10 0.533  Preprocesso~
## 12        4 sse_within_total standard   278.         10 5.10   Preprocesso~
## 13        5 sse_ratio      standard   0.189        10 0.00697 Preprocesso~
## 14        5 sse_total      standard 1227.         10 0.533  Preprocesso~
## 15        5 sse_within_total standard   232.         10 8.56   Preprocesso~
## 16        6 sse_ratio      standard   0.180        10 0.00745 Preprocesso~
## 17        6 sse_total      standard 1227.         10 0.533  Preprocesso~
## 18        6 sse_within_total standard   221.         10 9.12   Preprocesso~
## 19        7 sse_ratio      standard   0.154        10 0.00787 Preprocesso~
## 20        7 sse_total      standard 1227.         10 0.533  Preprocesso~
## 21        7 sse_within_total standard   189.         10 9.67   Preprocesso~
## 22        8 sse_ratio      standard   0.145        10 0.00714 Preprocesso~
## 23        8 sse_total      standard 1227.         10 0.533  Preprocesso~
## 24        8 sse_within_total standard   179.         10 8.75   Preprocesso~
## 25        9 sse_ratio      standard   0.124        10 0.00207 Preprocesso~
## 26        9 sse_total      standard 1227.         10 0.533  Preprocesso~
## 27        9 sse_within_total standard   152.         10 2.55   Preprocesso~
## 28       10 sse_ratio      standard   0.117        10 0.00144 Preprocesso~
## 29       10 sse_total      standard 1227.         10 0.533  Preprocesso~
## 30       10 sse_within_total standard   143.         10 1.77   Preprocesso~
```

```
## # A tibble: 1 x 2
##   num_clusters .config
##   <int> <chr>
## 1      10 Preprocessor1_Model10
```



A traves del metodo de silueta (silhouette)

```
##      bill_length_mm bill_depth_mm flipper_length_mm  body_mass_g
## [1,]    -0.88320467    0.78430007    -1.416271525  -0.563316704
## [2,]    -0.80993901    0.12600328    -1.060696087  -0.500969030
## [3,]    -0.66340769    0.42983257    -0.420660299  -1.186793445
## [4,]    -1.32279862    1.08812936    -0.562890474  -0.937402749
## [5,]    -0.84657184    1.74642615    -0.776235737  -0.688012052
## [6,]    -0.91983750    0.32855614    -1.416271525  -0.719185889
## [7,]    -0.86488825    1.24004400    -0.420660299   0.590115266
## [8,]    -1.79902541    0.48047078    -0.562890474  -0.906228912
## [9,]    -0.35202864    1.54387329    -0.776235737   0.060160036
## [10,]   -1.12131806   -0.02591137    -1.060696087  -1.124445771
## [11,]   -1.12131806    0.07536506    -1.487386613  -0.625664378
## [12,]   -0.51687637    0.22727971    -1.345156438  -1.249141119
## [13,]   -0.97478674    2.05025544    -0.705120649  -0.500969030
## [14,]   -1.70744334    1.99961722    -0.207315036   0.247203059
## [15,]   -1.34111504    0.32855614    -1.131811175  -0.625664378
## [16,]   -0.95647033    0.93621471    -0.420660299  -0.937402749
## [17,]   -0.26044656    1.79706436    -0.278430124   0.371898407
## [18,]   -1.74407616    0.63238542    -1.202926262  -1.093271934
## [19,]    0.38062795    2.20217008    -0.491775386  -0.002187638
## [20,]   -1.12131806    0.58174721    -1.914077138  -0.999750423
## [21,]   -1.13963448    0.78430007    -1.487386613  -0.750359726
```


##	[22,]	-1.46932994	1.03749114	-0.847350824	-0.500969030
##	[23,]	-1.04805240	0.48047078	-1.131811175	-0.313926008
##	[24,]	-0.93815391	0.02472685	-1.487386613	-0.500969030
##	[25,]	-1.57922843	0.88557650	-0.989581000	-0.500969030
##	[26,]	-0.60845845	0.73366185	-1.274041350	-0.812707400
##	[27,]	-0.62677486	0.37919435	-0.989581000	-1.249141119
##	[28,]	-1.10300165	0.73366185	-2.056307313	-1.311488793
##	[29,]	-0.62677486	0.88557650	-1.487386613	-0.313926008
##	[30,]	-0.80993901	-0.22846423	-1.629616788	-1.186793445
##	[31,]	-1.23121655	0.48047078	-1.629616788	-0.376273682
##	[32,]	-0.80993901	0.32855614	-0.918465912	-1.124445771
##	[33,]	-0.55350920	0.88557650	-1.202926262	-0.376273682
##	[34,]	-1.37774787	-0.07654958	-0.420660299	-1.093271934
##	[35,]	-0.86488825	1.99961722	-0.349545211	-0.064535312
##	[36,]	-0.93815391	1.44259686	-0.776235737	-0.313926008
##	[37,]	-0.31539581	0.68302364	-1.487386613	-0.812707400
##	[38,]	-1.15795089	1.08812936	-1.416271525	-1.124445771
##	[39,]	-0.75498976	0.98685293	-1.202926262	0.558941429
##	[40,]	-1.35943145	0.42983257	-1.345156438	-1.311488793
##	[41,]	-0.57182562	0.63238542	-0.420660299	-0.376273682
##	[42,]	-1.45101353	0.68302364	-1.060696087	-1.373836467
##	[43,]	0.03261607	1.29068222	-0.349545211	0.247203059
##	[44,]	-1.26784938	-0.12718780	-1.131811175	-1.498531815
##	[45,]	-0.79162259	0.83493828	-0.776235737	0.496593755
##	[46,]	-0.51687637	0.93621471	-1.345156438	-0.968576586
##	[47,]	-1.17626731	0.88557650	-1.558501700	-1.529705652
##	[48,]	-1.45101353	0.37919435	-0.776235737	-0.937402749
##	[49,]	-0.29707939	2.05025544	-0.705120649	-0.064535312
##	[50,]	-0.79162259	0.27791792	-1.060696087	-0.875055074
##	[51,]	-0.70004052	0.88557650	-0.918465912	0.122507710
##	[52,]	-1.63417768	0.37919435	-0.776235737	-0.937402749
##	[53,]	-0.35202864	1.18940579	-0.065084861	-0.189230660
##	[54,]	-1.72575975	0.48047078	-0.989581000	-1.623227163
##	[55,]	-0.46192713	0.73366185	-0.705120649	-0.625664378
##	[56,]	-0.90152108	0.17664149	-1.060696087	-0.812707400
##	[57,]	-0.60845845	0.83493828	-0.562890474	-0.500969030
##	[58,]	-1.35943145	-0.27910244	-1.416271525	-1.685574837
##	[59,]	-1.15795089	0.98685293	-0.491775386	-0.563316704
##	[60,]	-1.50596277	-0.12718780	-1.131811175	-1.311488793
##	[61,]	-0.48024354	1.99961722	-0.420660299	0.247203059
##	[62,]	-1.15795089	-0.07654958	-1.131811175	-0.750359726
##	[63,]	-0.51687637	0.53110900	-0.634005562	-0.189230660
##	[64,]	-1.37774787	-0.02591137	-1.202926262	-1.685574837
##	[65,]	-0.42529430	0.42983257	-0.634005562	-0.313926008
##	[66,]	-1.54259560	-0.48165530	-0.420660299	-1.062098097
##	[67,]	-0.51687637	0.98685293	-0.918465912	-0.126882986
##	[68,]	-1.46932994	-0.27910244	-0.776235737	-1.436184141
##	[69,]	-0.38866147	1.13876757	-0.207315036	0.309550733
##	[70,]	-1.90892390	0.93621471	-0.776235737	-0.750359726
##	[71,]	-0.77330618	0.63238542	-0.776235737	-0.376273682
##	[72,]	-0.79162259	0.02472685	-0.349545211	-0.812707400
##	[73,]	0.34399512	0.88557650	-0.278430124	-0.064535312
##	[74,]	-1.54259560	0.17664149	-0.776235737	-0.625664378
##	[75,]	-0.20549732	0.68302364	-0.420660299	0.060160036

## [76,]	-0.55350920	-0.17782601	-0.705120649	-0.625664378
## [77,]	-1.23121655	1.13876757	-1.202926262	-0.376273682
## [78,]	-1.41438070	-0.53229351	-0.989581000	-0.812707400
## [79,]	-0.33371222	0.98685293	-0.420660299	-0.251578334
## [80,]	-1.70744334	0.02472685	-0.847350824	-1.249141119
## [81,]	-0.18718091	0.22727971	-0.349545211	0.621289103
## [82,]	-1.32279862	0.83493828	-0.989581000	-0.500969030
## [83,]	-1.61586126	1.13876757	-0.562890474	-0.002187638
## [84,]	-1.21290014	0.32855614	-0.705120649	-1.062098097
## [85,]	-0.48024354	1.59451151	-0.491775386	-0.812707400
## [86,]	-1.39606428	1.18940579	-0.776235737	-0.500969030
## [87,]	-1.28616579	0.73366185	-0.847350824	-0.875055074
## [88,]	-1.02973599	1.03749114	-0.847350824	-0.313926008
## [89,]	-0.91983750	0.83493828	-0.776235737	-0.750359726
## [90,]	-1.50596277	0.42983257	0.077145314	-0.812707400
## [91,]	-0.51687637	0.48047078	0.290490577	0.122507710
## [92,]	-1.81734182	-0.02591137	-1.131811175	-0.999750423
## [93,]	-0.79162259	0.48047078	-1.060696087	0.309550733
## [94,]	-1.41438070	0.07536506	-0.989581000	-1.124445771
## [95,]	-0.57182562	0.88557650	0.503835840	0.122507710
## [96,]	-1.06636882	0.73366185	-0.776235737	-0.625664378
## [97,]	-0.66340769	0.68302364	-0.349545211	0.184855384
## [98,]	-1.98218956	-0.53229351	-1.629616788	-1.623227163
## [99,]	-0.13223166	0.68302364	-0.634005562	-0.126882986
## [100,]	-1.63417768	0.37919435	-0.634005562	-0.594490541
## [101,]	-0.53519279	1.44259686	0.148260402	0.652462940
## [102,]	-1.13963448	-0.58293173	-1.274041350	-1.405010304
## [103,]	-1.12131806	1.44259686	-0.776235737	0.060160036
## [104,]	-1.10300165	0.73366185	-0.562890474	-1.592053326
## [105,]	-0.77330618	0.88557650	-1.202926262	-0.812707400
## [106,]	-0.97478674	0.02472685	-0.136199948	-0.563316704
## [107,]	-1.04805240	1.44259686	-0.776235737	-0.376273682
## [108,]	-1.06636882	-0.07654958	-1.416271525	-1.280314956
## [109,]	-0.13223166	0.93621471	-0.278430124	0.714810614
## [110,]	-1.06636882	-0.32974066	-0.207315036	-0.469795193
## [111,]	0.30736229	1.59451151	-0.705120649	0.496593755
## [112,]	-0.77330618	0.27791792	-0.562890474	-1.249141119
## [113,]	-0.31539581	1.18940579	-0.278430124	0.091333873
## [114,]	-0.79162259	1.79706436	-0.705120649	-0.376273682
## [115,]	-0.22381374	0.58174721	-0.349545211	-0.158056823
## [116,]	-0.97478674	-0.07654958	-0.918465912	-1.623227163
## [117,]	-1.21290014	1.69578793	-0.136199948	-0.532142867
## [118,]	-1.50596277	-0.07654958	-0.847350824	-1.062098097
## [119,]	-0.51687637	0.73366185	-0.847350824	-1.093271934
## [120,]	-1.41438070	0.02472685	-0.989581000	-1.311488793
## [121,]	-1.13963448	1.34132043	-0.207315036	-0.875055074
## [122,]	-0.68172411	-0.07654958	-1.771846963	-0.937402749
## [123,]	-0.46192713	0.68302364	0.077145314	-0.407447519
## [124,]	-1.59754485	-0.63356994	-1.060696087	-1.436184141
## [125,]	-0.60845845	0.93621471	-0.136199948	-0.251578334
## [126,]	-0.93815391	0.22727971	-0.705120649	-1.155619608
## [127,]	-0.44361071	0.58174721	-0.420660299	0.122507710
## [128,]	-0.90152108	-0.02591137	-0.705120649	-1.436184141
## [129,]	0.03261607	0.42983257	0.646066015	-0.251578334

## [130,]	-0.99310316	0.37919435	-0.776235737	-1.093271934
## [131,]	-0.15054808	1.03749114	-0.278430124	-0.875055074
## [132,]	-1.30448221	0.68302364	-0.562890474	-0.875055074
## [133,]	-1.17626731	0.68302364	-0.136199948	0.340724570
## [134,]	-1.06636882	0.22727971	-0.989581000	-0.968576586
## [135,]	-0.51687637	0.17664149	-0.776235737	-0.376273682
## [136,]	-1.52427919	0.17664149	-0.705120649	-1.280314956
## [137,]	-0.68172411	1.49323508	-0.065084861	-0.282752171
## [138,]	-1.26784938	-0.32974066	-1.131811175	-0.999750423
## [139,]	-0.77330618	0.37919435	-0.562890474	0.060160036
## [140,]	-0.68172411	-0.02591137	-0.562890474	-0.999750423
## [141,]	-0.60845845	0.02472685	-0.989581000	-0.906228912
## [142,]	-2.16535371	-0.83612280	-0.918465912	-1.436184141
## [143,]	-0.59014203	-0.07654958	-0.776235737	-0.594490541
## [144,]	-1.21290014	-0.17782601	-0.634005562	-1.498531815
## [145,]	-0.90152108	0.78430007	-1.131811175	-0.688012052
## [146,]	-0.86488825	0.73366185	-0.776235737	0.060160036
## [147,]	-1.34111504	0.63238542	-1.202926262	-0.906228912
## [148,]	-1.45101353	0.32855614	-0.420660299	-0.937402749
## [149,]	-1.12131806	0.48047078	-0.562890474	-0.563316704
## [150,]	-1.45101353	-0.02591137	-0.989581000	-0.625664378
## [151,]	-0.44361071	0.68302364	0.006030227	-0.251578334
## [152,]	0.39894437	-2.00080174	0.717181103	0.371898407
## [153,]	1.11328455	-0.43101709	2.068367767	1.868242584
## [154,]	0.87517115	-1.54505781	0.646066015	0.309550733
## [155,]	1.11328455	-0.98803745	1.214986716	1.868242584
## [156,]	0.67369059	-1.34250495	1.001641453	1.494156540
## [157,]	0.47221003	-1.84888710	0.646066015	0.434246081
## [158,]	0.27072946	-1.29186674	0.717181103	0.745984451
## [159,]	0.50884286	-0.93739923	1.286101803	1.244765843
## [160,]	-0.11391525	-1.89952531	0.574950927	0.247203059
## [161,]	0.52715927	-0.88676102	1.001641453	1.182418169
## [162,]	-0.55350920	-1.74761067	0.930526365	0.558941429
## [163,]	0.93012040	-0.53229351	1.072756541	1.681199562
## [164,]	0.28904588	-1.74761067	0.930526365	0.558941429
## [165,]	0.82022191	-1.29186674	0.859411278	2.055285606
## [166,]	0.34399512	-1.29186674	0.646066015	-0.002187638
## [167,]	0.98506964	-0.73484637	1.143871628	2.055285606
## [168,]	-0.35202864	-1.84888710	0.646066015	-0.064535312
## [169,]	0.96675323	-0.98803745	1.428331979	2.616414673
## [170,]	0.41726078	-1.34250495	0.574950927	0.745984451
## [171,]	0.87517115	-1.03867566	1.499447066	1.431808866
## [172,]	1.14991738	-1.44378138	1.214986716	1.868242584
## [173,]	0.21578022	-1.34250495	1.001641453	0.995375147
## [174,]	0.47221003	-1.34250495	0.859411278	0.247203059
## [175,]	0.43557720	-0.68420816	1.001641453	1.057722821
## [176,]	-0.18718091	-2.05143996	1.001641453	0.995375147
## [177,]	0.39894437	-1.03867566	1.001641453	1.120070495
## [178,]	0.10588173	-1.44378138	1.072756541	-0.126882986
## [179,]	0.71032342	-1.08931388	1.001641453	1.805894910
## [180,]	0.78358908	-1.44378138	0.646066015	0.496593755
## [181,]	1.11328455	-0.93739923	1.357216891	1.681199562
## [182,]	0.61874135	-0.93739923	1.499447066	1.307113518
## [183,]	-0.20549732	-1.49441960	0.574950927	0.621289103

## [184,]	0.21578022	-1.34250495	0.432720752	1.057722821
## [185,]	2.87166037	-0.07654958	2.068367767	2.304676302
## [186,]	0.94843681	-1.19059031	1.357216891	1.182418169
## [187,]	0.82022191	-0.43101709	1.357216891	1.494156540
## [188,]	-0.24213015	-1.74761067	0.859411278	0.933027473
## [189,]	0.08756532	0.07536506	1.286101803	1.307113518
## [190,]	0.01429966	-1.79824888	0.503835840	0.184855384
## [191,]	0.87517115	-0.73484637	0.503835840	1.431808866
## [192,]	-0.22381374	-1.74761067	0.503835840	-0.313926008
## [193,]	1.04001889	-0.58293173	1.712792329	1.868242584
## [194,]	0.25241305	-1.74761067	0.646066015	0.122507710
## [195,]	1.04001889	-1.08931388	1.072756541	0.683636777
## [196,]	1.20486662	-0.63356994	1.499447066	1.681199562
## [197,]	-0.05896600	-1.64633424	1.143871628	0.870679799
## [198,]	0.28904588	-1.64633424	0.646066015	-0.002187638
## [199,]	1.20486662	-0.63356994	1.712792329	1.494156540
## [200,]	0.17914739	-1.95016353	0.859411278	1.120070495
## [201,]	0.23409663	-0.68420816	1.001641453	1.369461192
## [202,]	0.49052644	-1.49441960	0.646066015	0.808332125
## [203,]	0.83853832	-1.54505781	1.357216891	1.369461192
## [204,]	0.21578022	-1.39314317	0.646066015	0.247203059
## [205,]	1.13160096	-1.08931388	1.712792329	0.995375147
## [206,]	0.47221003	-1.39314317	1.143871628	0.870679799
## [207,]	0.19746381	-0.88676102	1.357216891	1.057722821
## [208,]	-0.02233317	-1.64633424	0.503835840	0.122507710
## [209,]	0.28904588	-1.08931388	1.357216891	0.995375147
## [210,]	-0.13223166	-1.34250495	0.503835840	0.309550733
## [211,]	1.18655021	-0.93739923	1.641677241	1.681199562
## [212,]	0.25241305	-1.69697245	0.503835840	-0.002187638
## [213,]	0.41726078	-1.13995209	1.428331979	1.369461192
## [214,]	0.32567871	-1.64633424	0.930526365	0.247203059
## [215,]	1.90089038	-0.73484637	2.139482854	1.805894910
## [216,]	0.34399512	-1.49441960	1.286101803	0.621289103
## [217,]	1.07665172	-0.17782601	2.068367767	1.868242584
## [218,]	0.41726078	-1.39314317	0.930526365	0.558941429
## [219,]	1.02170247	-0.48165530	1.997252679	1.992937932
## [220,]	-0.07728242	-1.49441960	1.357216891	0.621289103
## [221,]	1.24149945	-1.08931388	1.570562154	1.681199562
## [222,]	0.69200701	-1.08931388	1.072756541	0.683636777
## [223,]	0.45389361	-0.78548459	1.428331979	0.995375147
## [224,]	0.78358908	-0.78548459	1.428331979	1.120070495
## [225,]	0.47221003	-1.19059031	1.143871628	1.244765843
## [226,]	0.45389361	-1.08931388	1.072756541	0.621289103
## [227,]	0.85685474	-0.58293173	2.068367767	1.992937932
## [228,]	0.65537418	-1.49441960	0.574950927	0.496593755
## [229,]	1.31476511	-0.43101709	1.357216891	2.242328628
## [230,]	0.23409663	-1.69697245	1.001641453	0.683636777
## [231,]	0.23409663	-0.38037887	1.570562154	2.179980954
## [232,]	0.94843681	-1.34250495	0.788296190	0.527767592
## [233,]	1.57119492	-0.78548459	1.428331979	1.556504214
## [234,]	0.63705776	-1.29186674	0.788296190	0.652462940
## [235,]	1.11328455	-0.63356994	1.641677241	1.431808866
## [236,]	0.17914739	-1.69697245	0.788296190	0.683636777
## [237,]	1.25981586	0.07536506	1.926137592	1.743547236

## [238,]	-0.09559883	-1.39314317	1.214986716	0.496593755
## [239,]	1.35139794	-1.49441960	1.214986716	1.369461192
## [240,]	0.65537418	-1.59569603	0.788296190	0.839505962
## [241,]	1.49792926	-0.07654958	2.068367767	1.681199562
## [242,]	0.65537418	-1.08931388	1.214986716	0.933027473
## [243,]	1.51624567	-0.02591137	1.926137592	1.494156540
## [244,]	0.28904588	-1.34250495	0.788296190	0.683636777
## [245,]	1.02170247	-0.53229351	1.641677241	1.805894910
## [246,]	0.10588173	-1.24122852	0.930526365	0.808332125
## [247,]	1.25981586	-0.73484637	1.783907417	1.244765843
## [248,]	1.00338606	-0.68420816	1.072756541	0.901853636
## [249,]	0.54547569	-1.29186674	1.499447066	0.839505962
## [250,]	0.82022191	-1.39314317	0.148260402	0.527767592
## [251,]	1.31476511	-0.32974066	1.712792329	1.307113518
## [252,]	0.83853832	-1.08931388	1.286101803	0.808332125
## [253,]	2.19395302	-0.07654958	1.926137592	1.743547236
## [254,]	0.60042493	-0.83612280	1.001641453	0.964201310
## [255,]	0.94843681	-1.08931388	1.926137592	1.618851888
## [256,]	0.61874135	-1.69697245	1.072756541	0.652462940
## [257,]	0.52715927	-0.53229351	1.001641453	1.618851888
## [258,]	-0.40697788	-1.24122852	0.646066015	0.621289103
## [259,]	1.73604265	-0.68420816	1.286101803	1.618851888
## [260,]	-0.11391525	-1.59569603	0.503835840	0.465419918
## [261,]	0.76527266	-1.03867566	0.574950927	1.618851888
## [262,]	1.20486662	-0.98803745	1.072756541	0.995375147
## [263,]	1.07665172	-0.63356994	1.997252679	2.179980954
## [264,]	-0.07728242	-0.98803745	0.859411278	0.558941429
## [265,]	1.38803077	-0.43101709	2.068367767	1.618851888
## [266,]	0.41726078	-1.54505781	1.143871628	0.216029222
## [267,]	2.04742170	-0.58293173	2.068367767	2.055285606
## [268,]	0.10588173	-0.73484637	1.143871628	0.839505962
## [269,]	0.89348757	-0.48165530	1.499447066	2.242328628
## [270,]	0.60042493	-1.74761067	0.930526365	0.901853636
## [271,]	0.52715927	-1.44378138	1.001641453	0.808332125
## [272,]	1.18655021	-0.73484637	1.499447066	1.930590258
## [273,]	0.23409663	-1.19059031	0.788296190	1.244765843
## [274,]	1.09496813	-0.53229351	0.859411278	1.494156540
## [275,]	0.47221003	0.37919435	-0.634005562	-0.875055074
## [276,]	1.11328455	1.18940579	-0.349545211	-0.376273682
## [277,]	1.35139794	1.03749114	-0.562890474	-0.688012052
## [278,]	0.27072946	0.78430007	-0.918465912	-0.843881237
## [279,]	1.60782775	1.34132043	-0.278430124	-0.594490541
## [280,]	0.23409663	0.32855614	-0.207315036	-0.313926008
## [281,]	0.39894437	0.53110900	-1.629616788	-1.186793445
## [282,]	1.35139794	0.53110900	-0.278430124	-0.563316704
## [283,]	0.38062795	0.88557650	-0.420660299	-0.064535312
## [284,]	1.35139794	1.39195865	-0.207315036	-0.625664378
## [285,]	0.49052644	0.32855614	-0.562890474	-0.500969030
## [286,]	1.42466360	1.59451151	-0.491775386	-0.532142867
## [287,]	0.56379210	0.07536506	-1.131811175	-0.625664378
## [288,]	1.47961284	0.48047078	0.006030227	-0.189230660
## [289,]	0.36231154	-0.02591137	-0.776235737	-0.781533563
## [290,]	1.20486662	1.24004400	0.006030227	-0.189230660
## [291,]	1.16823379	1.44259686	-0.278430124	-1.124445771

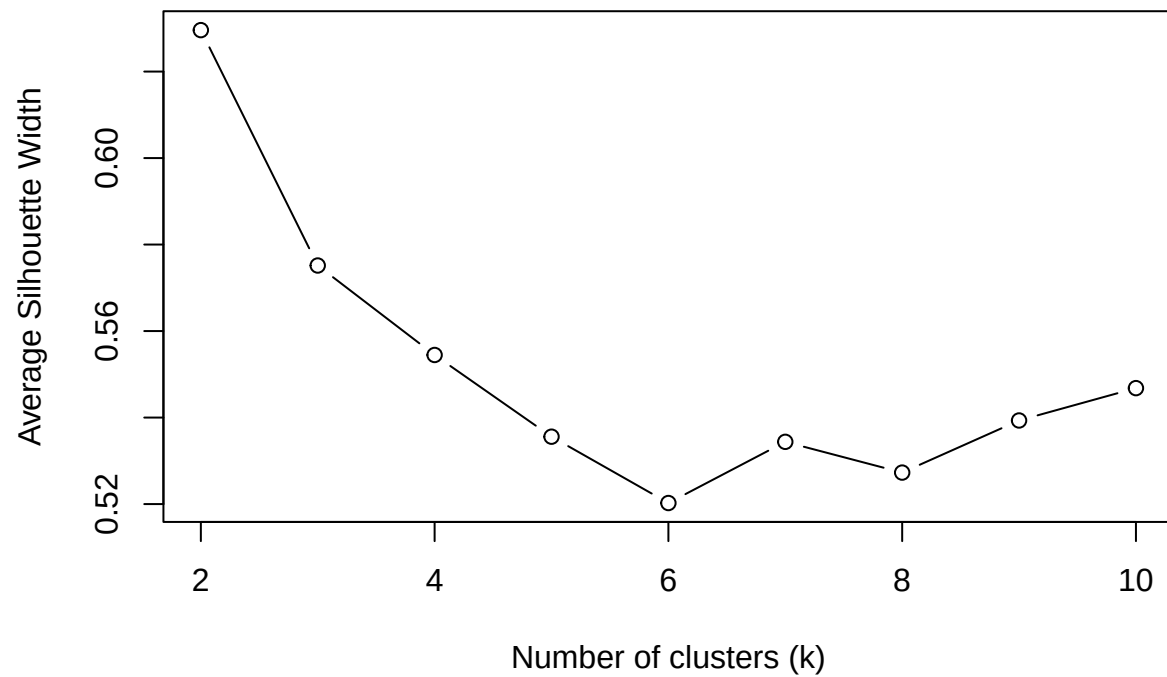
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## [292,]      2.57859773      0.32855614     -1.416271525 -0.625664378
## [293,]      0.45389361      0.73366185     -0.776235737 -0.937402749
## [294,]      0.96675323      0.53110900     -0.420660299  0.247203059
## [295,]     -0.27876298      0.07536506     -1.416271525 -0.750359726
## [296,]      0.83853832      0.17664149     -0.705120649 -0.999750423
## [297,]     -0.13223166     -0.27910244     -0.989581000 -1.623227163
## [298,]      1.22318303      1.13876757     -0.562890474 -0.500969030
## [299,]      0.50884286      0.37919435     -0.420660299 -1.124445771
## [300,]      1.47961284      0.93621471     -0.278430124 -0.064535312
## [301,]      1.20486662      0.63238542     -0.065084861 -0.999750423
## [302,]      1.02170247      0.93621471     -0.065084861 -0.500969030
## [303,]      0.45389361      0.32855614     -0.705120649 -0.625664378
## [304,]      1.62614416      1.44259686      0.290490577  0.434246081
## [305,]     -0.55350920     -0.27910244     -0.989581000 -1.249141119
## [306,]      1.88257397      1.84770258      0.006030227  0.122507710
## [307,]     -0.26044656     -0.22846423     -0.989581000 -1.062098097
## [308,]      1.29644869      0.83493828      0.148260402 -0.126882986
## [309,]      1.05833530      0.73366185     -0.420660299 -0.750359726
## [310,]      0.65537418     -0.17782601     -0.136199948 -0.376273682
## [311,]      0.67369059      0.58174721     -0.420660299 -0.438621356
## [312,]      1.47961284      1.79706436      0.646066015  0.745984451
## [313,]      0.54547569     -0.27910244     -0.634005562 -1.872617859
## [314,]      1.75435906      1.39195865      0.290490577  0.371898407
## [315,]      0.93012040      1.18940579      0.646066015 -0.313926008
## [316,]      0.41726078      0.17664149     -0.989581000 -0.688012052
## [317,]      1.27813228      0.98685293     -0.349545211 -0.812707400
## [318,]      0.28904588     -0.07654958     -0.349545211 -0.875055074
## [319,]      1.27813228      0.37919435     -0.349545211 -0.656838215
## [320,]      1.25981586      0.68302364      0.006030227  0.309550733
## [321,]      1.13160096      0.37919435     -0.776235737 -0.999750423
## [322,]      0.93012040      1.24004400      0.788296190  0.122507710
## [323,]      1.38803077      0.78430007     -0.989581000 -1.186793445
## [324,]      1.07665172      0.07536506     -0.207315036 -0.656838215
## [325,]      0.76527266     -0.38037887     -0.136199948 -1.093271934
## [326,]      1.36971435      0.93621471      0.006030227 -0.313926008
## [327,]      0.32567871      0.07536506     -0.562890474 -0.750359726
## [328,]      1.24149945      1.29068222      0.148260402 -0.189230660
## [329,]     -0.26044656      0.07536506     -0.989581000 -1.062098097
## [330,]      1.51624567      0.83493828     -0.278430124 -0.937402749
## [331,]      0.23409663     -0.27910244     -0.705120649 -1.186793445
## [332,]      0.98506964      1.39195865      0.148260402 -0.189230660
## [333,]      1.14991738      0.83493828      0.077145314 -0.500969030
## [334,]      0.30736229      1.13876757     -0.491775386 -0.843881237
## [335,]      1.46129643      1.18940579      0.361605665 -0.313926008
## [336,]      0.52715927     -0.32974066     -0.847350824 -0.688012052
## [337,]      0.32567871     -0.07654958     -0.420660299 -0.688012052
## [338,]      2.17563660      1.34132043      0.432720752 -0.251578334
## [339,]     -0.07728242      0.48047078      0.077145314 -0.999750423
## [340,]      1.04001889      0.53110900     -0.562890474 -0.532142867
## [341,]      1.25981586      0.93621471      0.646066015 -0.126882986
## [342,]      1.14991738      0.78430007     -0.207315036 -0.532142867
## attr(,"scaled:center")
##      bill_length_mm      bill_depth_mm flipper_length_mm      body_mass_g
##              43.92193              17.15117             200.91520             4201.75439

```

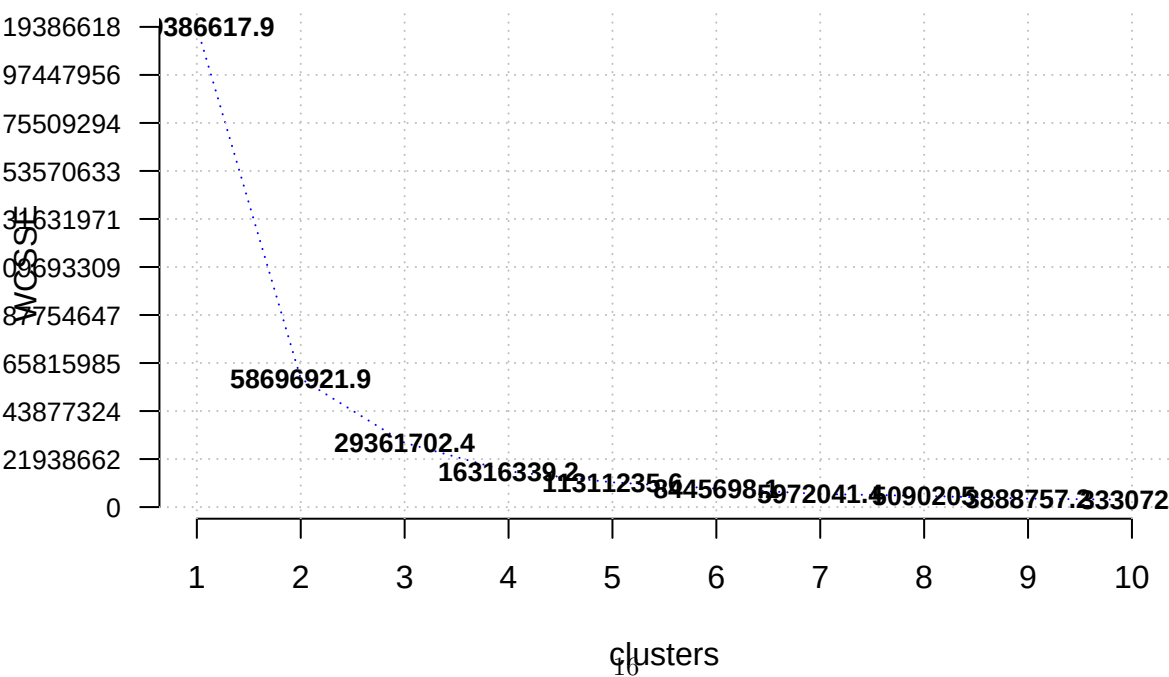
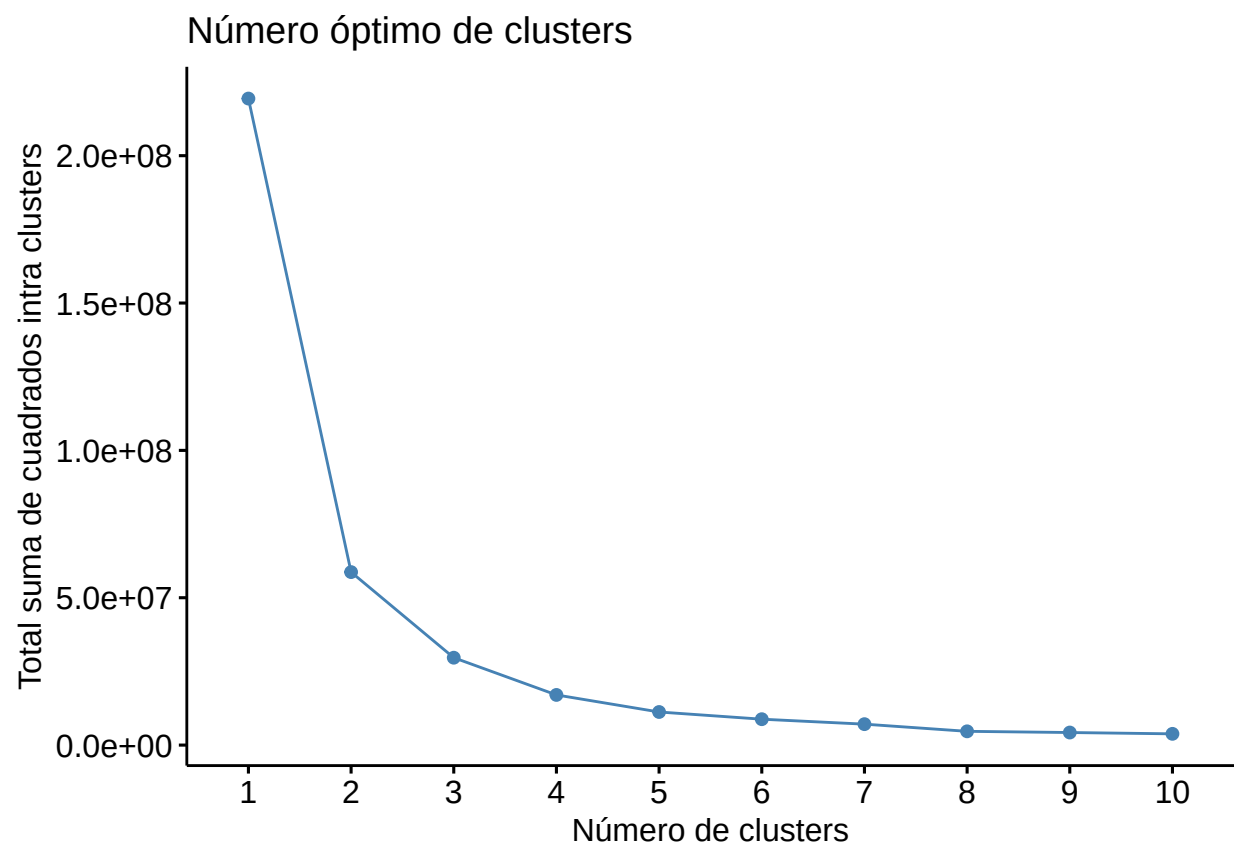
```
## attr("scaled:scale")
##   bill_length_mm  bill_depth_mm flipper_length_mm  body_mass_g
##      5.459584      1.974793      14.061714      801.954536
```

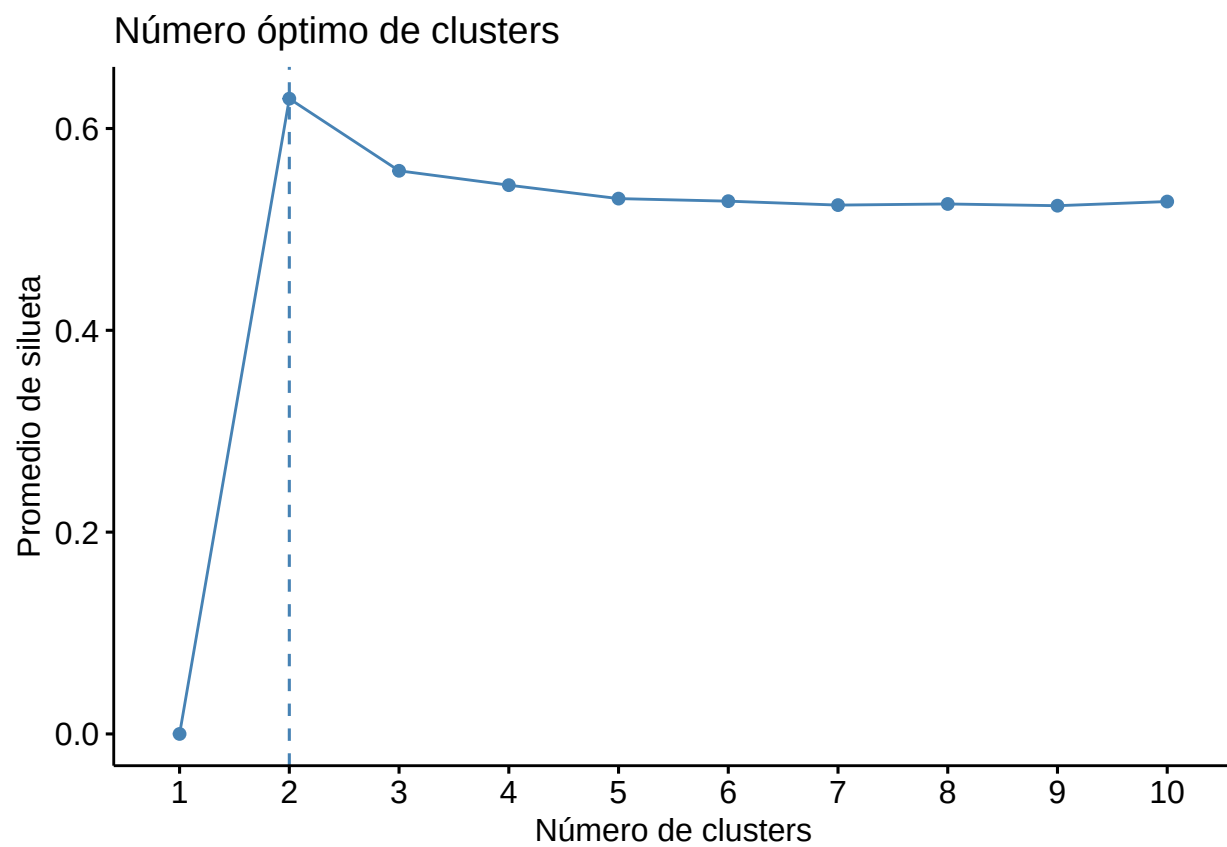
Silhouette Method for Optimal k

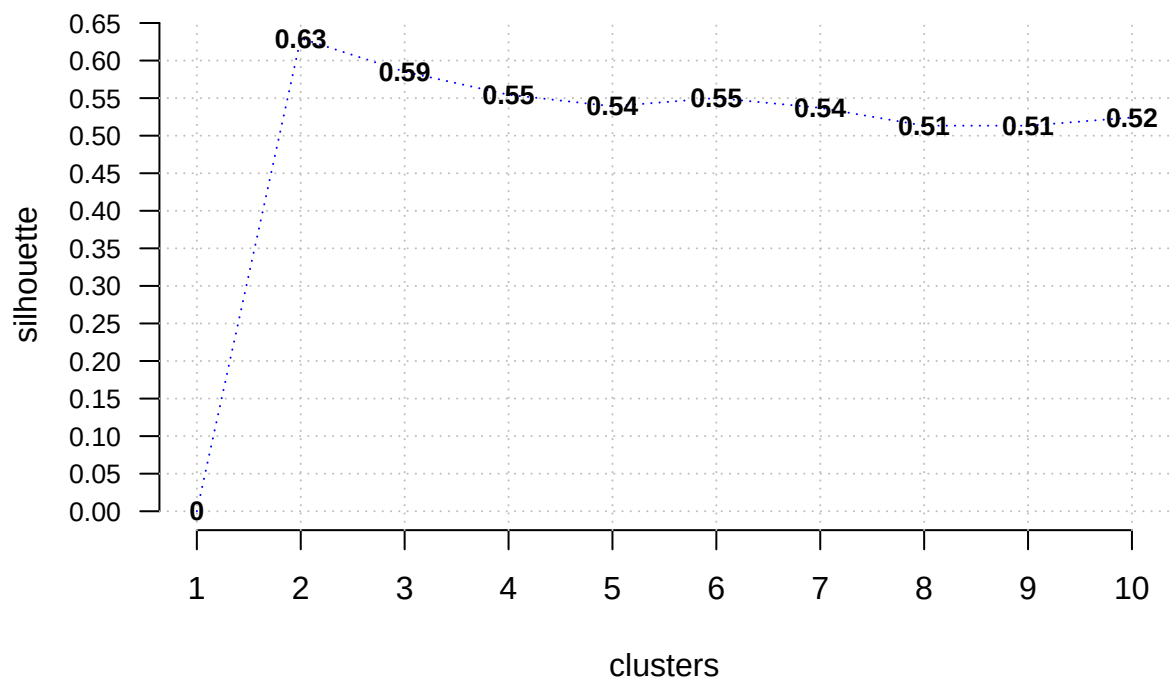


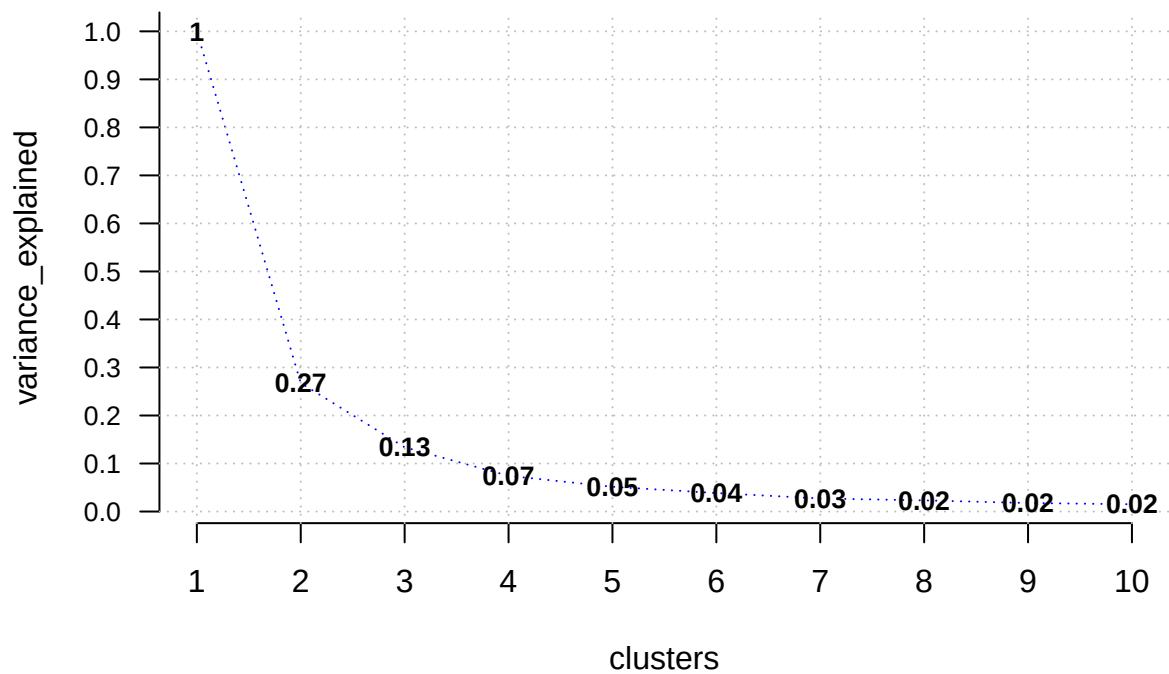
```
## [1] "Optimal number of clusters (k): 2"
```

Explorando el numero de clustes K subyacentes a estos datos con otros metodos

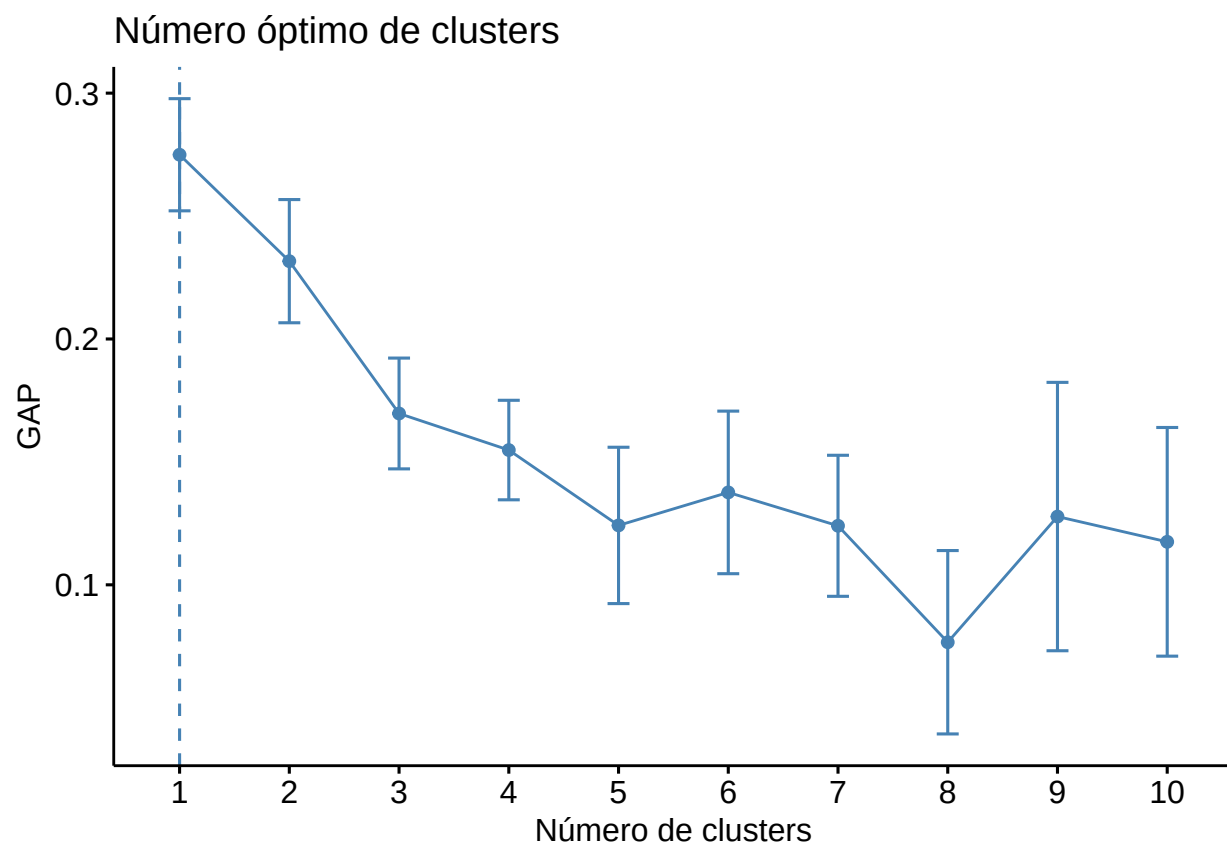


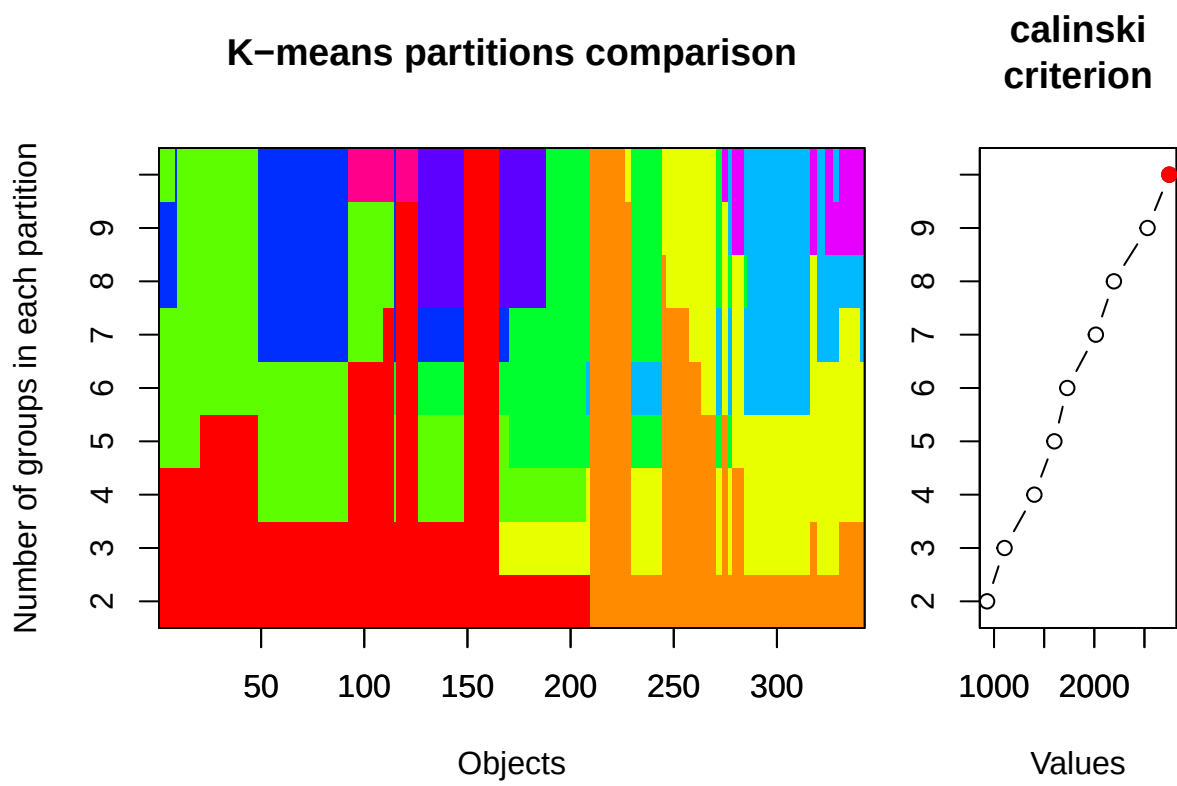


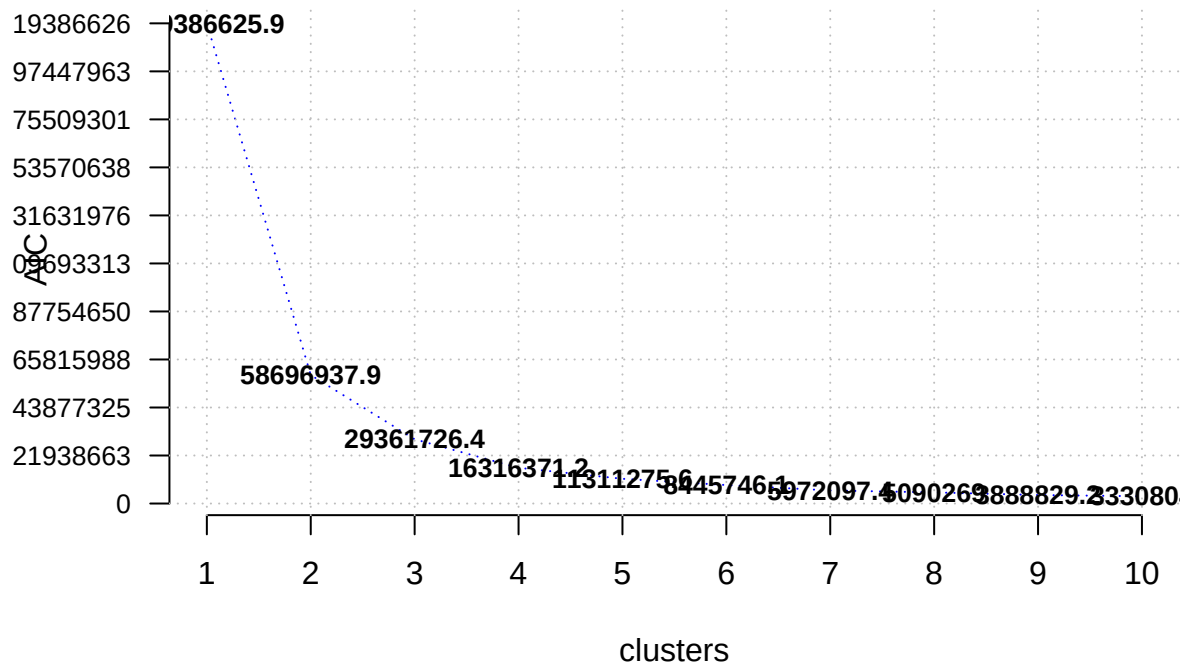


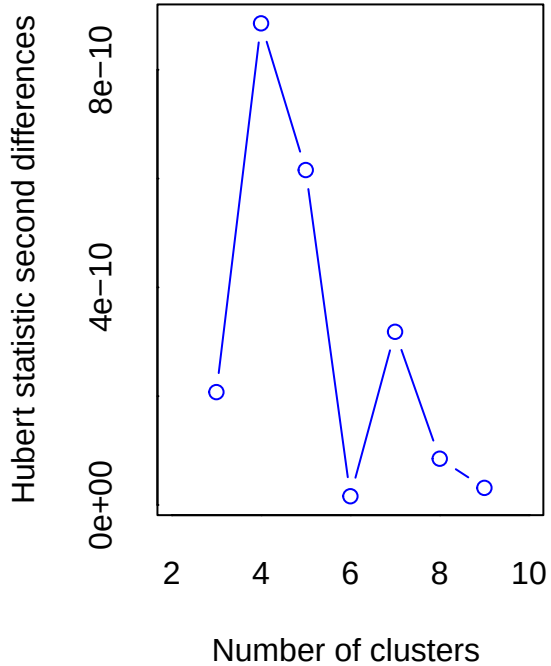
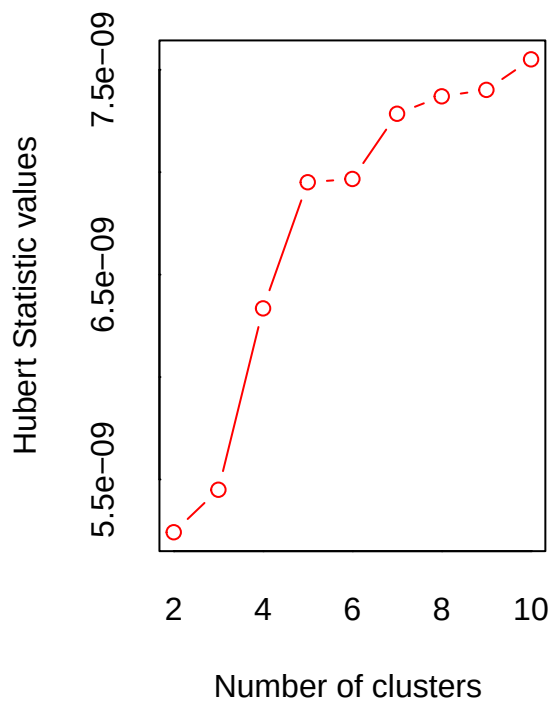


```
## Clustering k = 1,2,..., K.max (= 10): .. done
## Bootstrapping, b = 1,2,..., B (= 100) [one "." per sample]:
## ..... 50
## ..... 100
```

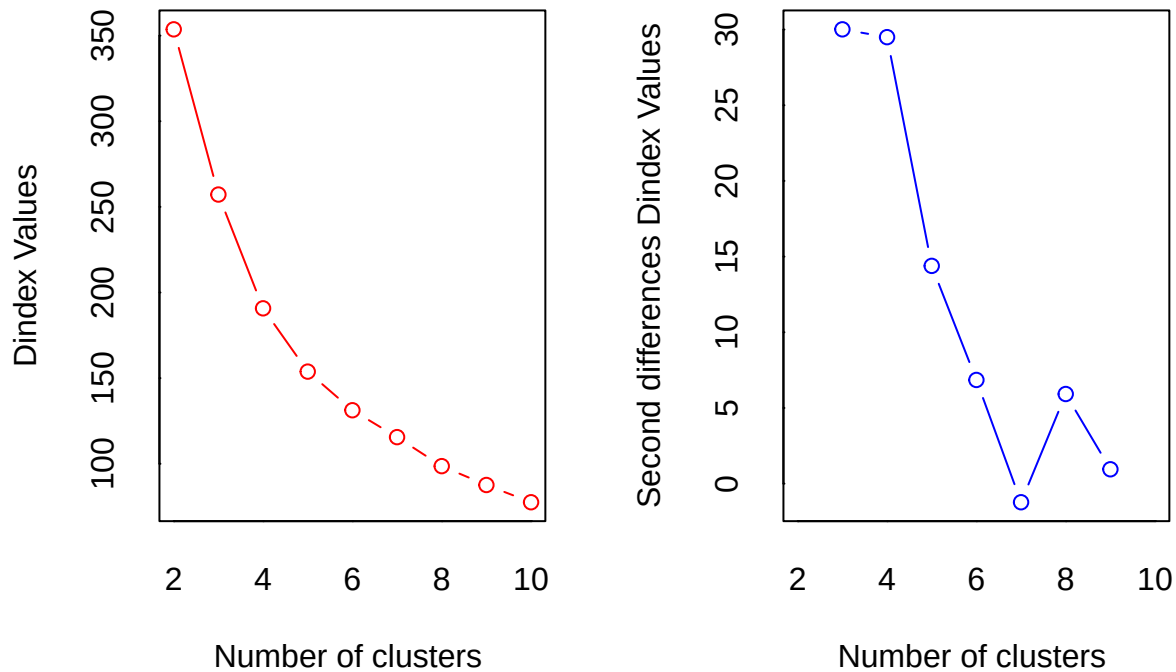








```
## *** : The Hubert index is a graphical method of determining the number of clusters.
##       In the plot of Hubert index, we seek a significant knee that corresponds to a
##       significant increase of the value of the measure i.e the significant peak in Hubert
##       index second differences plot.
##
```



```
## *** : The D index is a graphical method of determining the number of clusters.
##           In the plot of D index, we seek a significant knee (the significant peak in Dindex
##           second differences plot) that corresponds to a significant increase of the value of
##           the measure.
##
## *****
## * Among all indices:
## * 7 proposed 2 as the best number of clusters
## * 3 proposed 3 as the best number of clusters
## * 2 proposed 4 as the best number of clusters
## * 3 proposed 5 as the best number of clusters
## * 1 proposed 7 as the best number of clusters
## * 5 proposed 10 as the best number of clusters
##
##           ***** Conclusion *****
##
## * According to the majority rule, the best number of clusters is  2
##
## *****

## [1] "All.index"          "All.CriticalValues" "Best.nc"
## [4] "Best.partition"

##           KL           CH Hartigan       CCC       Scott       Marriot
```



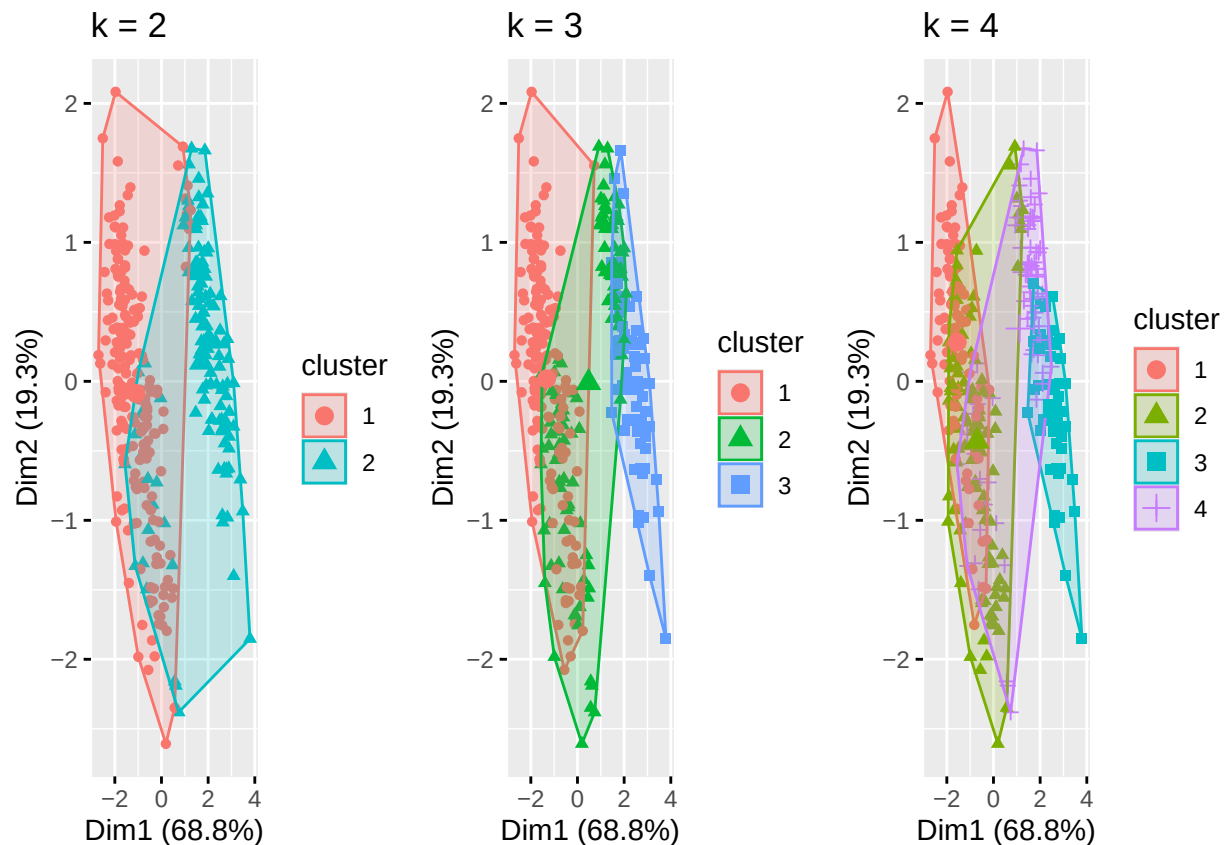
```
## Number_clusters 2.0000 10.000 5.0000 2.0000 4.0000 5.00000e+00
## Value_Index      5.0442 2414.772 102.5553 59.7595 213.2073 2.34927e+18
##               TrCovW  TraceW Friedman      Rubin  Cindex      DB
## Number_clusters 3.000000e+00      3 10.0000 5.0000 10.0000 10.0000
## Value_Index      6.175077e+14 10753210 491.3776 -51.5444 0.1218 0.5763
##               Silhouette Duda PseudoT2 Beale Ratkowsky      Ball PtBiserial
## Number_clusters 2.0000 NA      NA      NA      2.0000      3      2.0000
## Value_Index      0.6265 NA      NA      NA      0.4822 18896261      0.7083
##               Frey McClain  Dunn Hubert SDindex Dindex      SDbw
## Number_clusters 2.0000 2.000 7.0000      0 4.0000      0 10.0000
## Value_Index      1.7461 0.271 0.0317      0 0.0051      0 0.1171
```

¿2, 3, o 4, clusters?

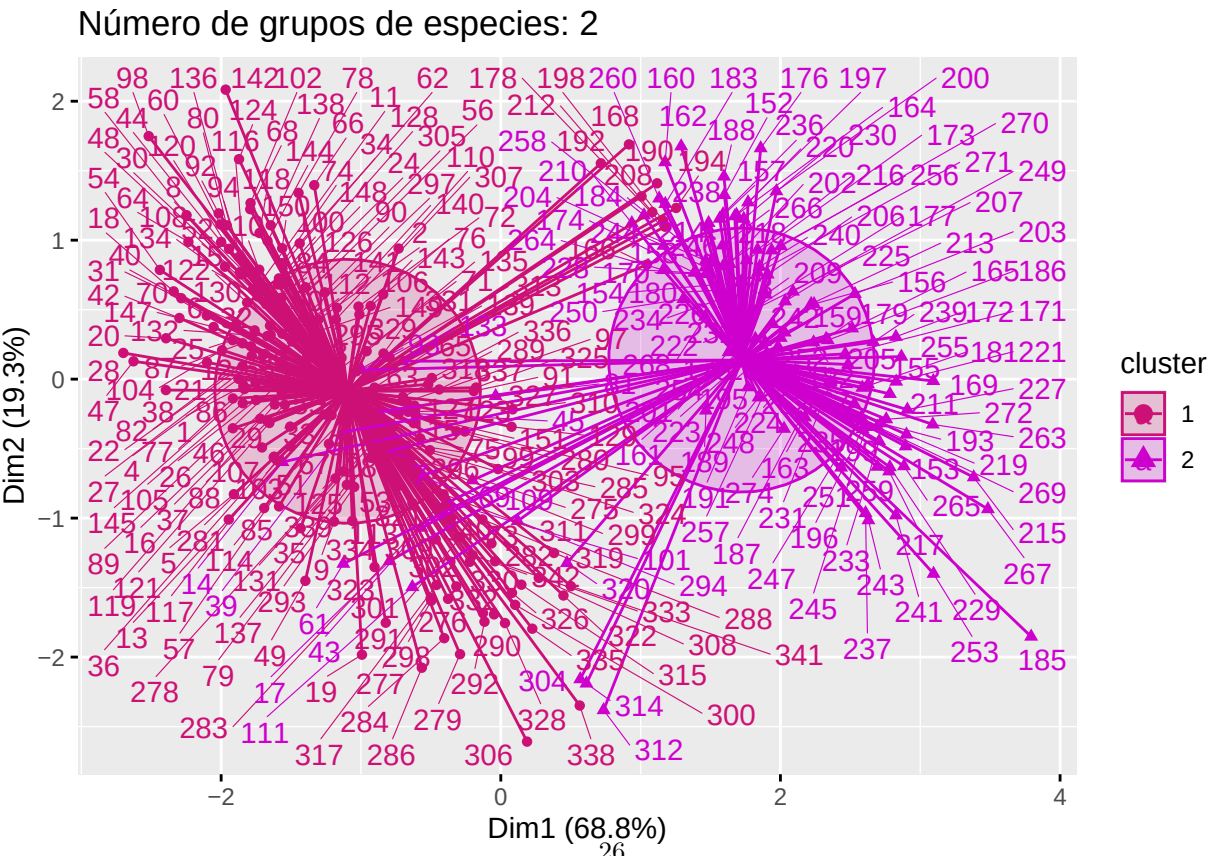
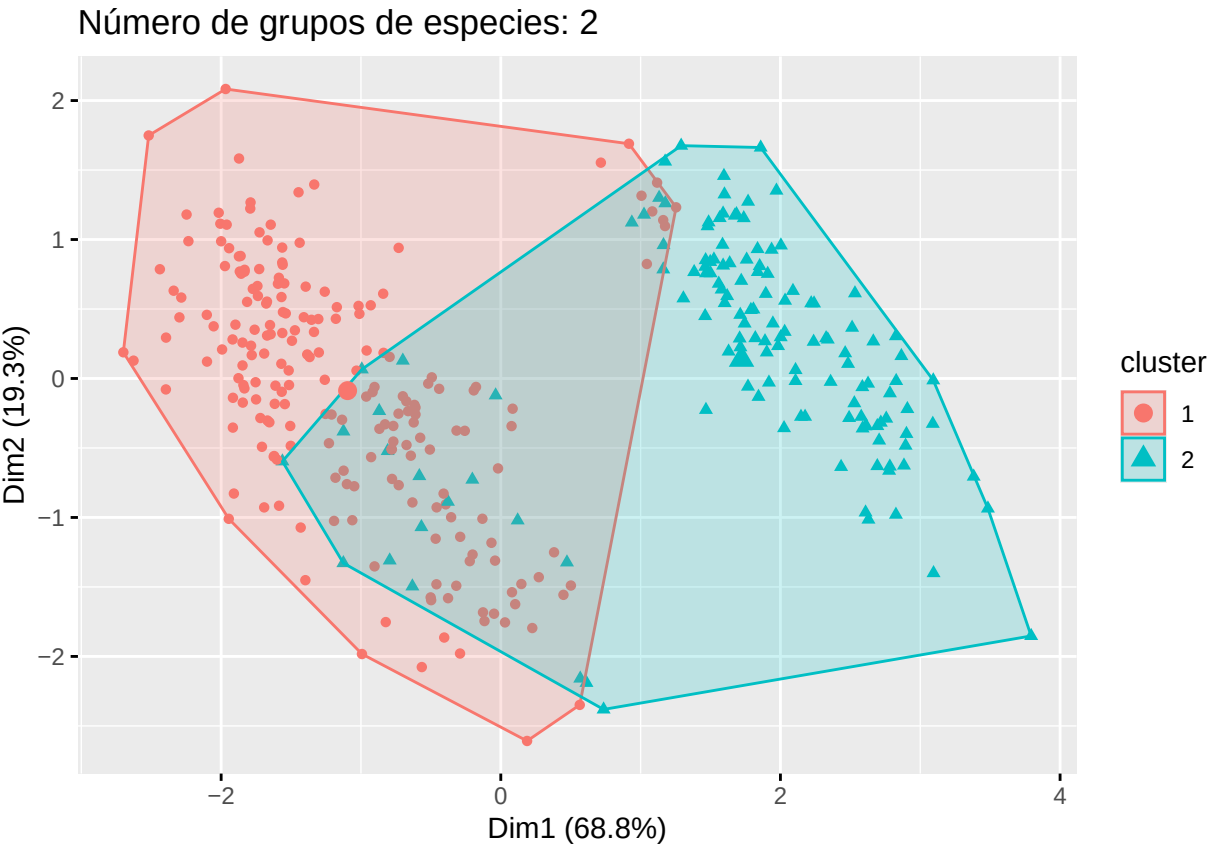
```
## [1] 0.2675501
```

```
## [1] 0.1329996
```

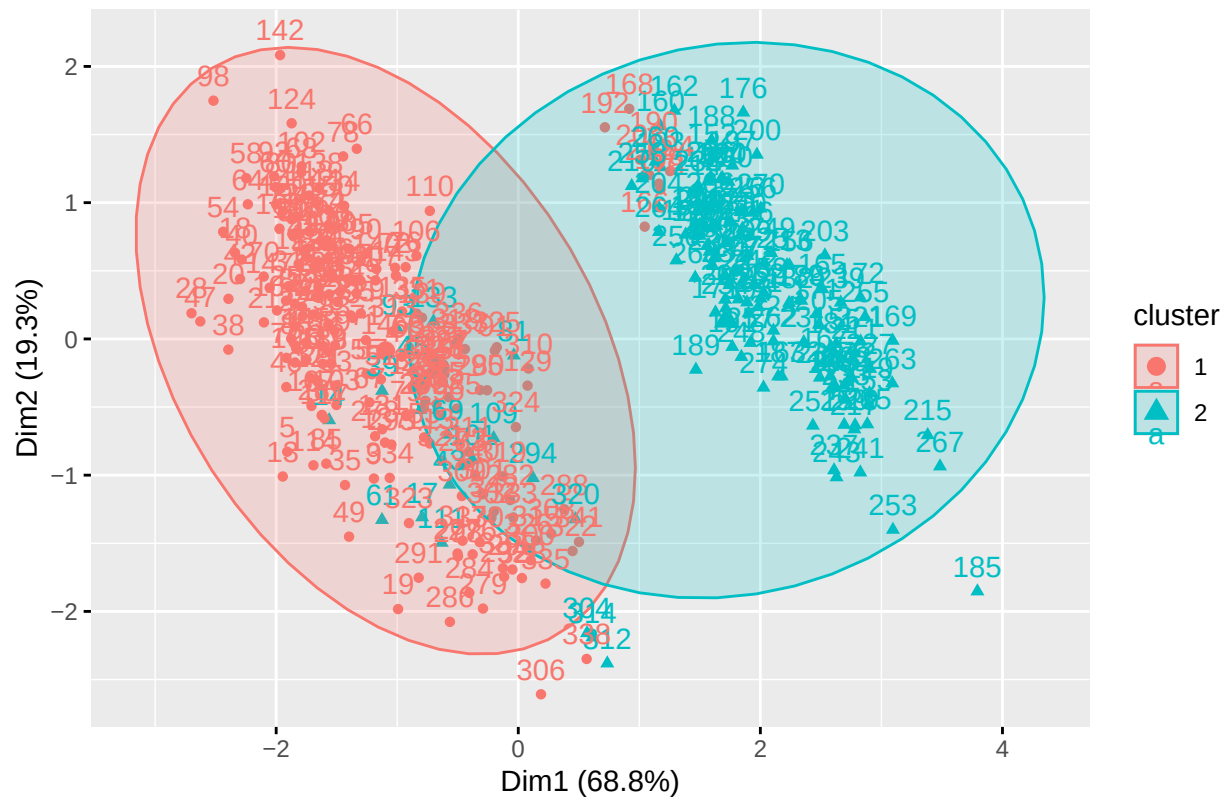
```
## [1] 0.07437254
```

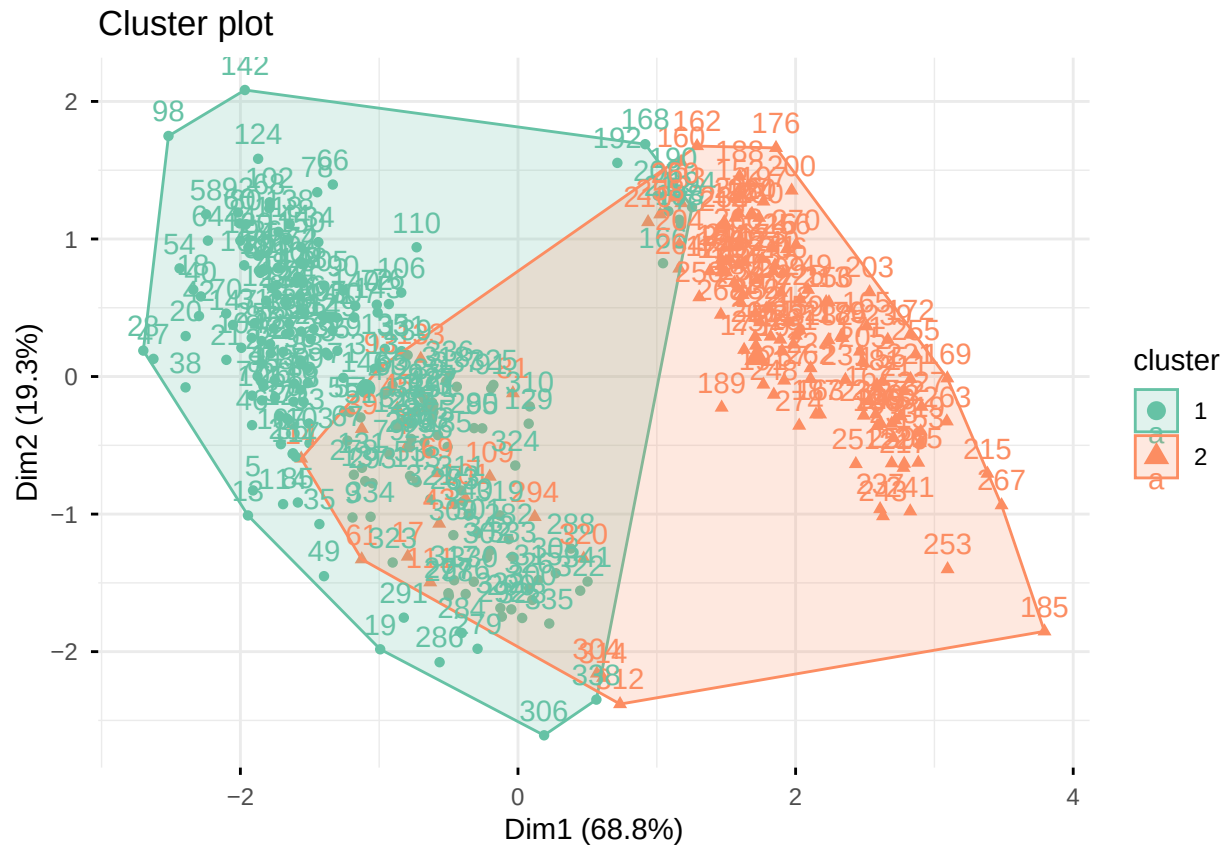


Concluimos que hay $K=2$ grupos de especies

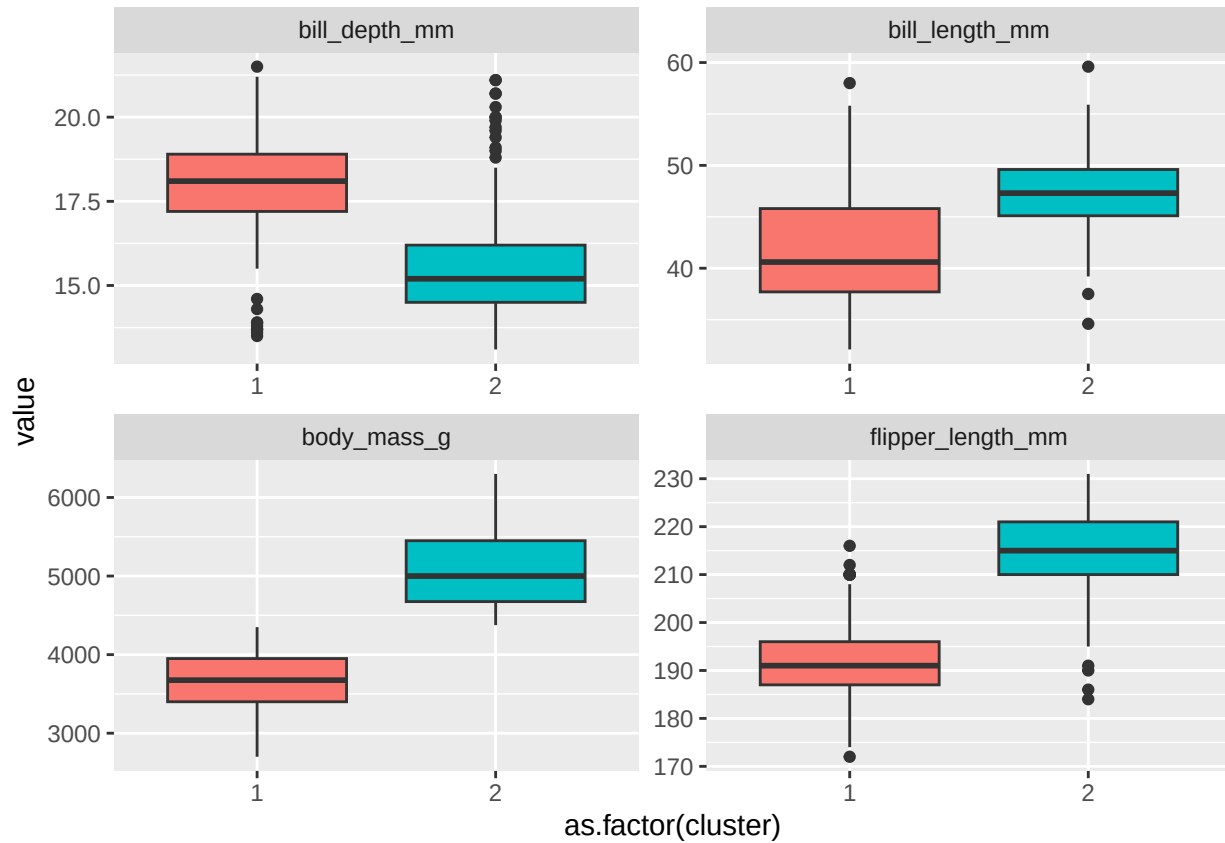


Cluster plot





```
## # A tibble: 2 x 5
##   .cluster bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>      <dbl>         <dbl>         <dbl>         <dbl>
## 1 Cluster_1      41.8           18.1           192.         3655.
## 2 Cluster_2      47.2           15.7           215.         5061.
```



```
## # A tibble: 2 x 3
##   cluster num_users pct_users
##   <int>     <int>     <dbl>
## 1       1       209     0.611
## 2       2       133     0.389
```

```
##
##           1  2
## Adelie    137 14
## Chinstrap  63  5
## Gentoo     9 114
```

```
## [1] 0.4152047
```

Hacemos cluster, primero haciendo reduccion de dimension a traves de PCA

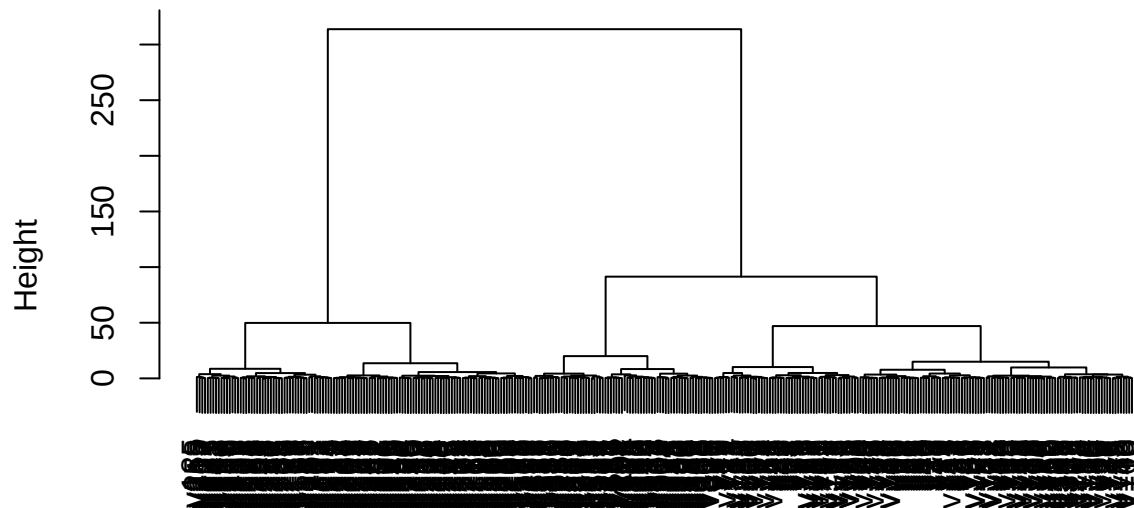
```
## [1] 0.9728769
```

```
## [1] 1
```

Clusters herarquicos

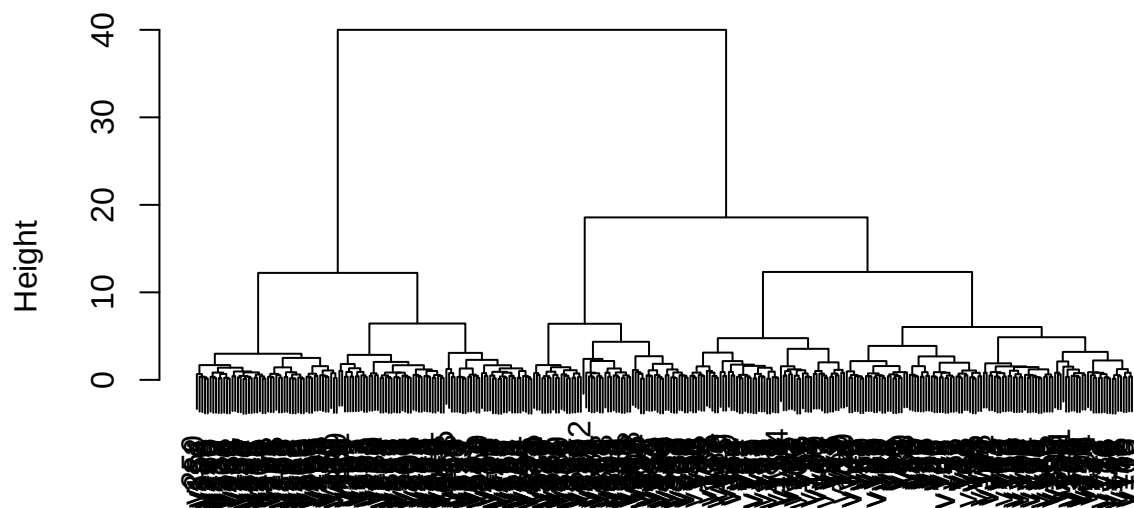
Cluster herarquicos trae pca incluido

Cluster Dendrogram



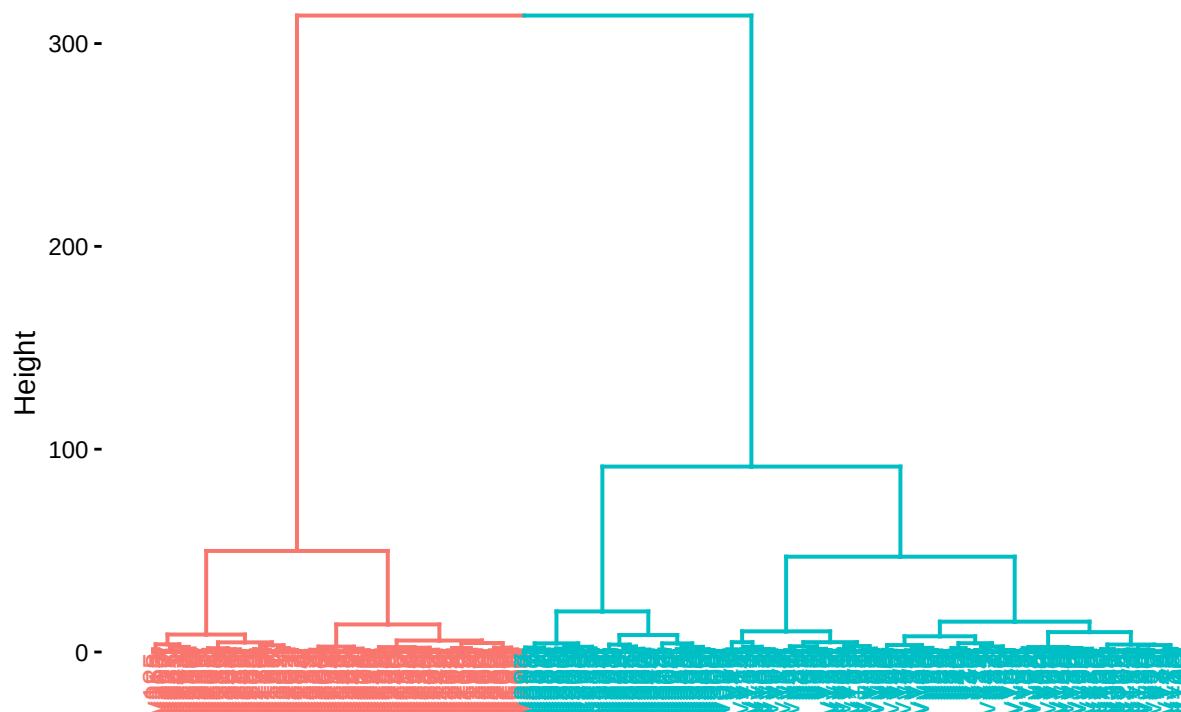
```
stats::as.dist(dmat)  
stats::hclust (*, "ward.D")
```

Cluster Dendrogram



```
30  
stats::as.dist(dmat)  
stats::hclust (*, "ward.D2")
```

Dendrograma basado en PCA: Liga Ward



K-Means

```
## K Means Cluster Specification (partition)
##
## Main Arguments:
##   num_clusters = 2
##
## Computational engine: ClusterR

## == Workflow [trained] =====
## Preprocessor: Recipe
## Model: k_means()
##
## -- Preprocessor -----
## 2 Recipe Steps
##
## * step_normalize()
## * step_pca()
##
## -- Model -----
## KMeans Cluster
## Call: ClusterR::KMeans_rcpp(data = data, clusters = clusters, num_init = num_init,      max_iters =
## Data cols: 4
## Centroids: 2
```

```
## BSS/SS: 0.586471
## SS: 1364 = 564.0535 (WSS) + 799.9465 (BSS)
```

```
## == Workflow =====
## Preprocessor: Recipe
## Model: k_means()
##
## -- Preprocessor -----
## 2 Recipe Steps
##
## * step_normalize()
## * step_pca()
##
## -- Model -----
## K Means Cluster Specification (partition)
##
## Main Arguments:
##   num_clusters = tune()
##
## Computational engine: ClusterR
```

```
## # A tibble: 20 x 7
```

	num_clusters	.metric	.estimator	mean	n	std_err	.config
	<int>	<chr>	<chr>	<dbl>	<int>	<dbl>	<chr>
## 1	6	sse_total	standard	1364	10	1.96e-13	Preprocessor1_~
## 2	6	sse_within_total	standard	235.	10	1.05e+ 1	Preprocessor1_~
## 3	9	sse_total	standard	1364	10	2.02e-13	Preprocessor1_~
## 4	9	sse_within_total	standard	172.	10	9.50e+ 0	Preprocessor1_~
## 5	3	sse_total	standard	1364	10	2.13e-13	Preprocessor1_~
## 6	3	sse_within_total	standard	408.	10	1.54e+ 1	Preprocessor1_~
## 7	2	sse_total	standard	1364	10	1.89e-13	Preprocessor1_~
## 8	2	sse_within_total	standard	549.	10	5.52e+ 0	Preprocessor1_~
## 9	7	sse_total	standard	1364	10	2.01e-13	Preprocessor1_~
## 10	7	sse_within_total	standard	207.	10	9.07e+ 0	Preprocessor1_~
## 11	5	sse_total	standard	1364	10	1.98e-13	Preprocessor1_~
## 12	5	sse_within_total	standard	261.	10	8.46e+ 0	Preprocessor1_~
## 13	10	sse_total	standard	1364	10	2.02e-13	Preprocessor1_~
## 14	10	sse_within_total	standard	150.	10	3.16e+ 0	Preprocessor1_~
## 15	4	sse_total	standard	1364	10	1.98e-13	Preprocessor1_~
## 16	4	sse_within_total	standard	295.	10	3.25e+ 0	Preprocessor1_~
## 17	1	sse_total	standard	1364	10	2.02e-13	Preprocessor1_~
## 18	1	sse_within_total	standard	1364	10	2.02e-13	Preprocessor1_~
## 19	8	sse_total	standard	1364	10	1.93e-13	Preprocessor1_~
## 20	8	sse_within_total	standard	190.	10	1.01e+ 1	Preprocessor1_~

```
## # A tibble: 342 x 1
##   .cluster
##   <fct>
## 1 Cluster_1
## 2 Cluster_1
## 3 Cluster_1
## 4 Cluster_1
## 5 Cluster_1
```



```
## 6 Cluster_1
## 7 Cluster_1
## 8 Cluster_1
## 9 Cluster_1
## 10 Cluster_1
## 11 Cluster_1
## 12 Cluster_1
## 13 Cluster_1
## 14 Cluster_1
## 15 Cluster_1
## 16 Cluster_1
## 17 Cluster_1
## 18 Cluster_1
## 19 Cluster_1
## 20 Cluster_1
## 21 Cluster_1
## 22 Cluster_1
## 23 Cluster_1
## 24 Cluster_1
## 25 Cluster_1
## 26 Cluster_1
## 27 Cluster_1
## 28 Cluster_1
## 29 Cluster_1
## 30 Cluster_1
## 31 Cluster_1
## 32 Cluster_1
## 33 Cluster_1
## 34 Cluster_1
## 35 Cluster_1
## 36 Cluster_1
## 37 Cluster_1
## 38 Cluster_1
## 39 Cluster_1
## 40 Cluster_1
## 41 Cluster_1
## 42 Cluster_1
## 43 Cluster_1
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## 47 Cluster_1
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## 51 Cluster_1
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## 54 Cluster_1
## 55 Cluster_1
## 56 Cluster_1
## 57 Cluster_1
## 58 Cluster_1
## 59 Cluster_1
```

```
## 60 Cluster_1
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## 111 Cluster_1
## 112 Cluster_1
## 113 Cluster_1
```

```
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## 124 Cluster_1
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## 150 Cluster_1
## 151 Cluster_1
## 152 Cluster_2
## 153 Cluster_2
## 154 Cluster_2
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## 167 Cluster_2
```

168 Cluster_2
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```
## 222 Cluster_2
## 223 Cluster_2
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## 272 Cluster_2
## 273 Cluster_2
## 274 Cluster_2
## 275 Cluster_1
```

276 Cluster_1
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```

## 330 Cluster_1
## 331 Cluster_1
## 332 Cluster_1
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## 335 Cluster_1
## 336 Cluster_1
## 337 Cluster_1
## 338 Cluster_1
## 339 Cluster_1
## 340 Cluster_1
## 341 Cluster_1
## 342 Cluster_1

## # A tibble: 2 x 5
##   .cluster    PC1      PC2      PC3      PC4
##   <fct>      <dbl> <dbl> <dbl> <dbl>
## 1 Cluster_1 -1.12 -0.219 -0.0205 -0.00292
## 2 Cluster_2  2.00  0.390  0.0365  0.00519

## # A tibble: 10 x 1
##   .pred_cluster
##   <fct>
## 1 Cluster_1
## 2 Cluster_1
## 3 Cluster_2
## 4 Cluster_2
## 5 Cluster_1
## 6 Cluster_1
## 7 Cluster_2
## 8 Cluster_1
## 9 Cluster_1
## 10 Cluster_1

## $cluster_names
## [1] Cluster_1 Cluster_2
## Levels: Cluster_1 Cluster_2
##
## $centroids
## # A tibble: 2 x 4
##   PC1      PC2      PC3      PC4
##   <dbl> <dbl> <dbl> <dbl>
## 1 -1.12 -0.219 -0.0205 -0.00292
## 2  2.00  0.390  0.0365  0.00519
##
## $n_members
## [1] 219 123
##
## $sse_within_total_total
## [1] 420.9033 143.1502
##
## $sse_total
## [1] 1364

```

[illegible]


```

## [281] Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1
## [288] Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1
## [295] Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1
## [302] Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1
## [309] Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1
## [316] Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1
## [323] Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1
## [330] Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1
## [337] Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1
## Levels: Cluster_1 Cluster_2

## # A tibble: 342 x 2
##   orig_labels standard_labels
##   <int> <fct>
## 1         1 Cluster_1
## 2         1 Cluster_1
## 3         1 Cluster_1
## 4         1 Cluster_1
## 5         1 Cluster_1
## 6         1 Cluster_1
## 7         1 Cluster_1
## 8         1 Cluster_1
## 9         1 Cluster_1
## 10        1 Cluster_1
## # i 332 more rows

##
##   Cluster_1 Cluster_2
## 1       219         0
## 2         0       123

##
##           1  2
## Adelie    151  0
## Chinstrap  68  0
## Gentoo     0 123

```

Comparacion de clasificacion por K-Means: Todas las variables vs. CP

```

## [1] Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1 Cluster_1
## Levels: Cluster_1 Cluster_2

##   class_pca
##   Cluster_1 Cluster_2
## 1       200         9
## 2        19       114

## # A tibble: 1 x 3
##   .metric      .estimator .estimate
##   <chr>        <chr>        <dbl>
## 1 silhouette_avg standard      0.532

```

Graficas finales

```
## # A tibble: 342 x 5
##   species    PC1      PC2      PC3      PC4
##   <fct>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 Adelie  -1.84 -0.0476  0.232  0.523
## 2 Adelie  -1.30  0.428   0.0295  0.402
## 3 Adelie  -1.37  0.154  -0.198 -0.527
## 4 Adelie  -1.88  0.00205  0.618 -0.478
## 5 Adelie  -1.91 -0.828   0.686 -0.207
## 6 Adelie  -1.76  0.351  -0.0276  0.504
## 7 Adelie  -0.809 -0.522   1.33   0.338
## 8 Adelie  -1.83  0.769   0.689 -0.427
## 9 Adelie  -1.19 -1.02    0.729  0.333
## 10 Adelie -1.73  0.787  -0.205  0.0205
## # i 332 more rows
```

